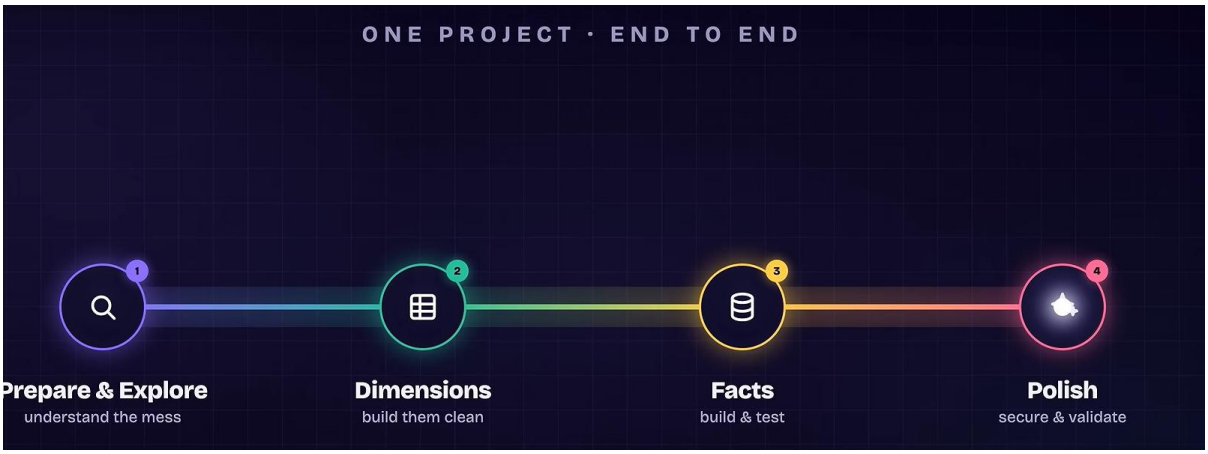
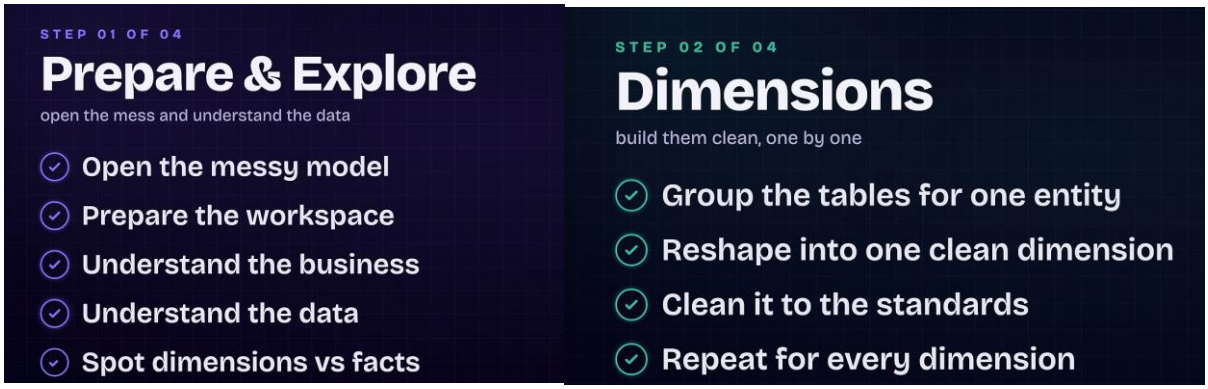
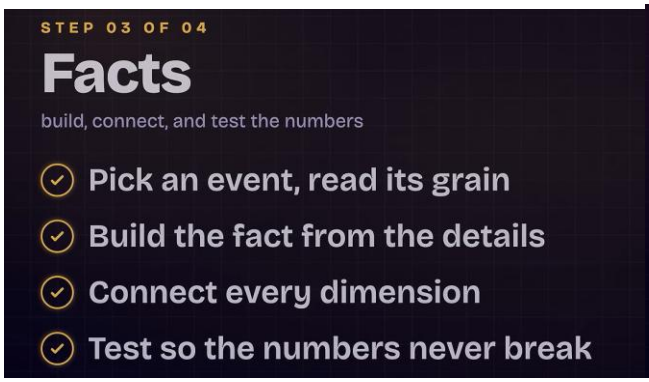
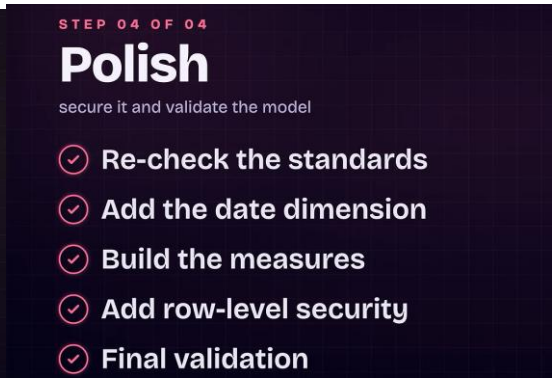
DATA MODELLING PROJECT

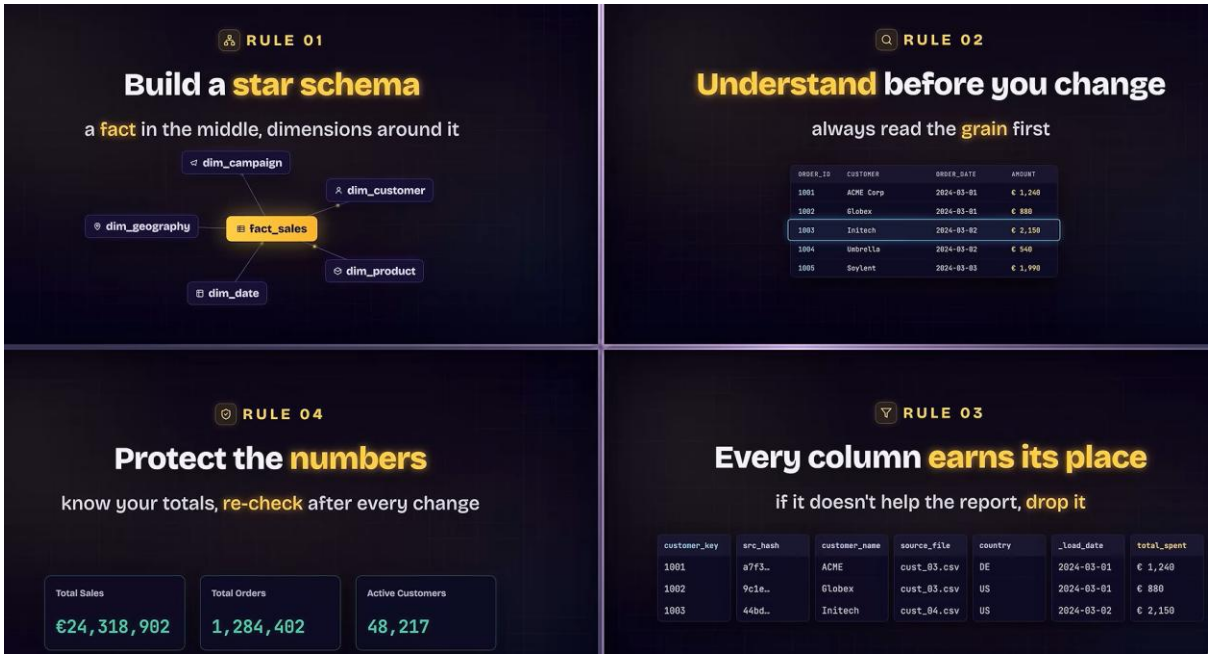






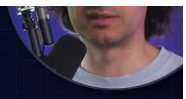


RULES TO KEEP IN MIND



The Standards

of our data modeling project



LANGUAGE	NAMING	TABLES	KEYS	FRIENDLY
English	snake_case	fact_ dim_	_key _id	readable
customer_name	customer_name	fact_sales	customer_key	customer_name
order_date	order_date	dim_customer	customer_id	order_status
ship_country	total_sales	dim_product	product_key	total_revenue

STANDARD · LANGUAGE

one language everywhere: English

land ⟶ country ✓

bestell_date ⟶ order_date ✓

gesamt_sales ⟶ total_sales ✓

kunden_name ⟶ customer_name ✓

STANDARD · NAMING

snake_case

all lowercase, joined with _

CustomerName ⟶ customer_name ✓

Order_Date ⟶ order_date ✓

total-sales ⟶ total_sales ✓

shipcountry ⟶ ship_country ✓

STANDARD · TABLES

fact_ for fact tables

dim_ for dimension tables

sales ⟶ fact_sales ✓

inventory ⟶ fact_inventory ✓

d_customer ⟶ dim_customer ✓

dimension_product ⟶ dim_product ✓

STANDARD · KEYS

_key = we make it
_id = from the source

id_customer →	customer_id ✓
CustKey →	customer_key ✓
UID-prd →	product_key ✓
order_pk →	order_id ✓

STANDARD · FRIENDLY NAMES

readable names, not cryptic codes

cust_nm →	customer_name ✓
tot_rev →	total_revenue ✓
ord_st_cd →	order_status ✓
shp_ctry →	ship_country ✓

THE SETUP



the order
PHASES

1. Prepare & Explore
2. Dimensions
3. Facts
4. Polish

the mindset
RULES

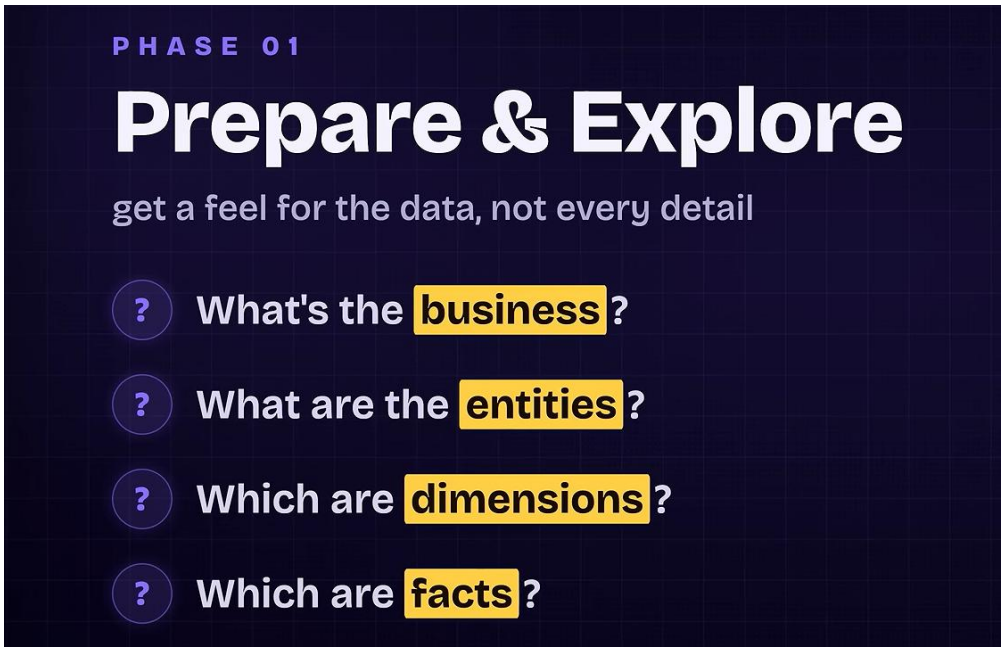
- Build a star schema
- Read the grain
- Earn its place
- Protect the numbers

the spec
STANDARDS

language	English
naming	snake_case
tables	fact_ / dim_
keys	_key / _id
names	friendly

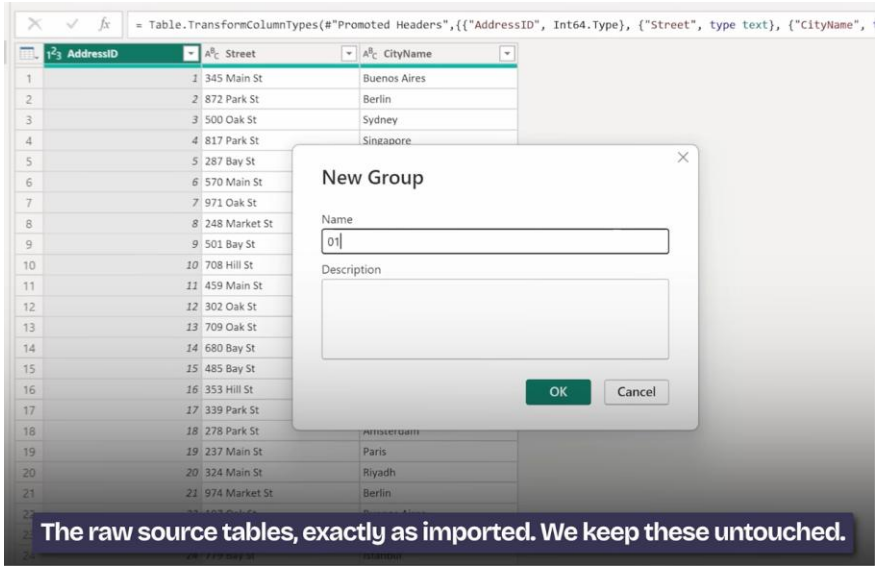
PROJECT IMPLEMENTATION

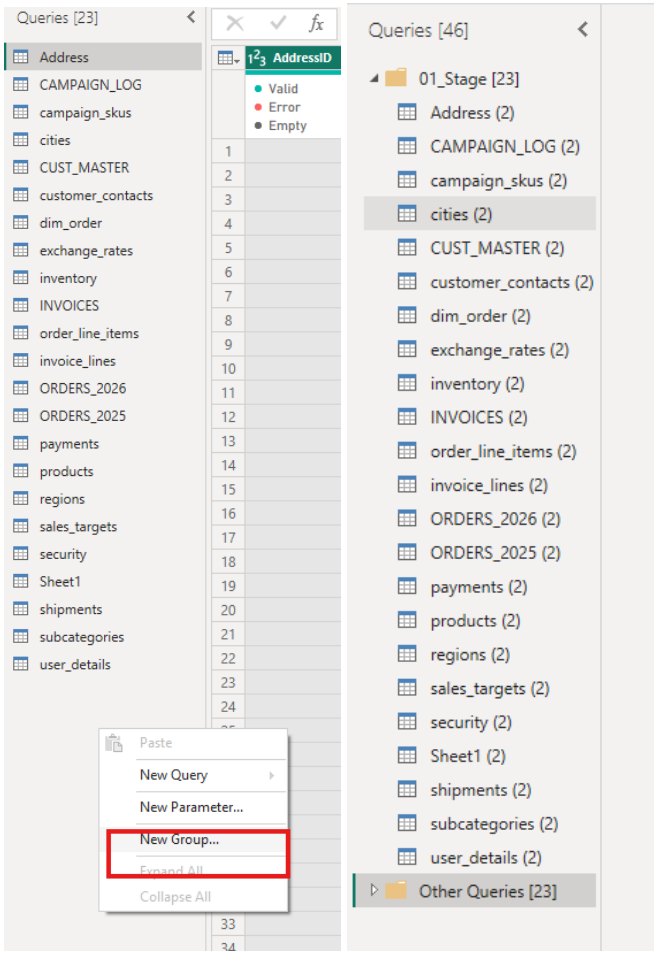
Phase 1



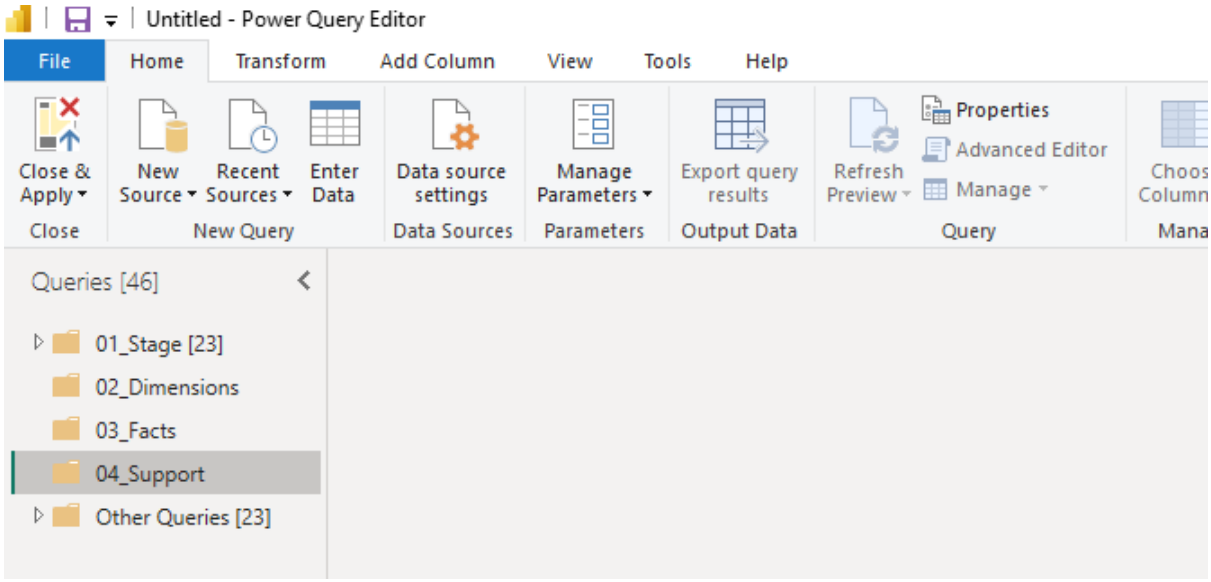


Before start preparing data model, organize the tables in power Query first. Open power Query and prepare a new folders for facts, dimensions and helps / support for tables which are not used in fact or dimensions

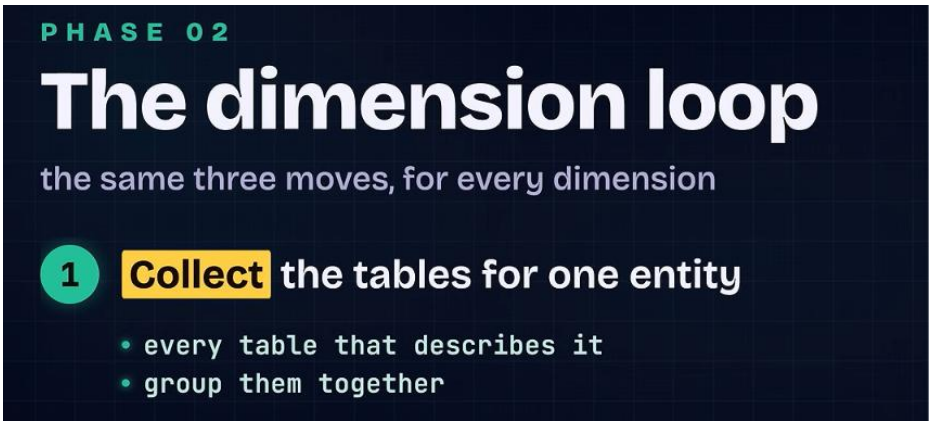


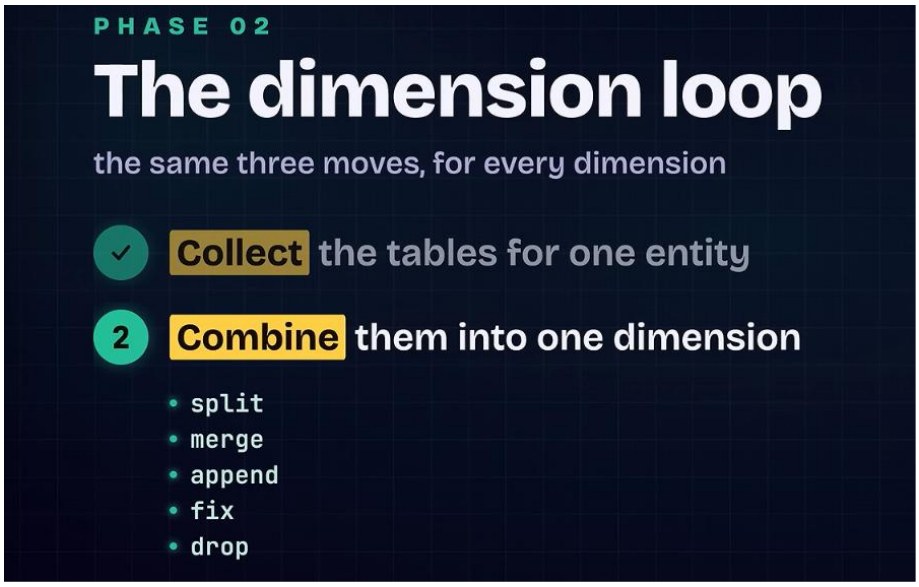


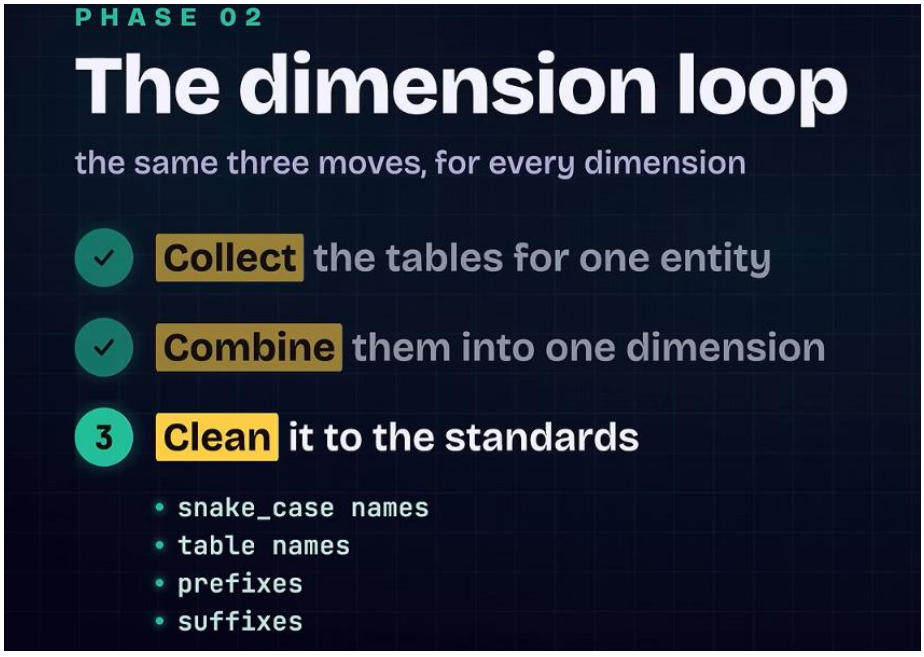
Same way Prepare New Group for ***Dimensions , Facts and helper or support tables that are neither fact nor dimensions***



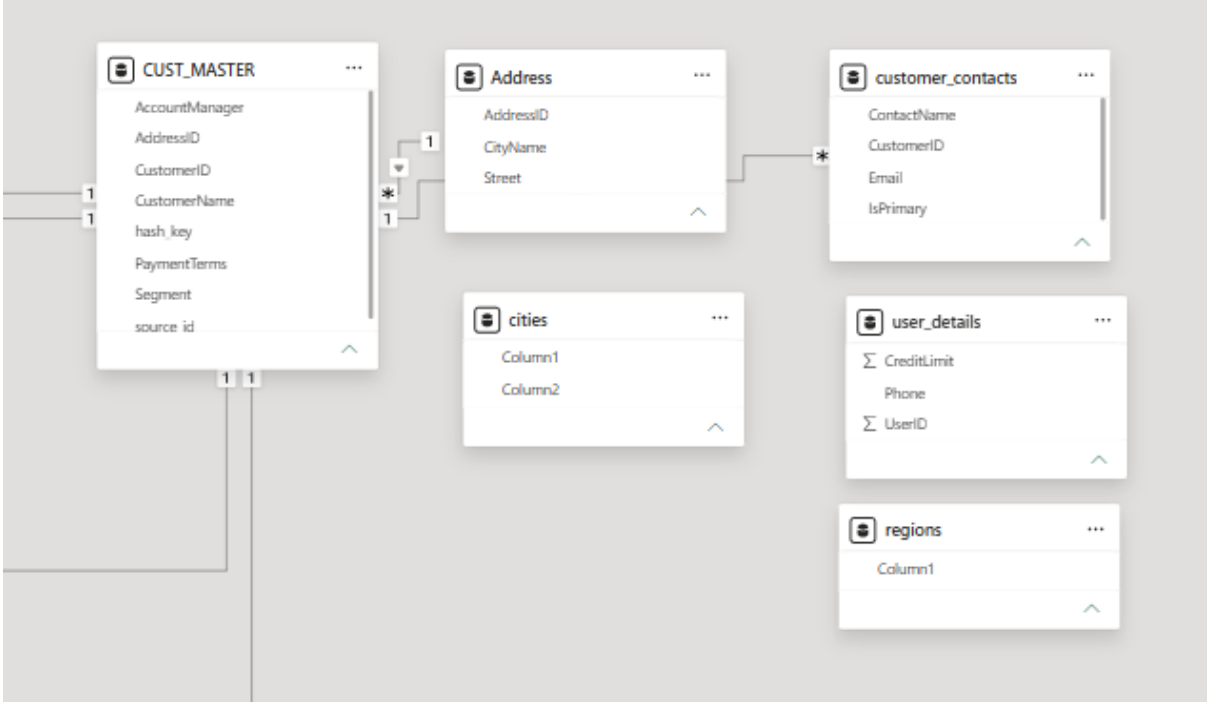
Phase 2 –Building Dimensions







- *Pick one Topic or entity and go ahead and start preparing dimensions*
- *Pick all the tables in data model that are related to the the picked topic or entity and turn them into one dimension.*
- *In this documentation we started with **customers topic***



- *Lets turn all those table into one dimension*
- *Please make sure when merging tables together grains are same once merged, if not it will lead to duplicate and other issues.*
- *After doing merging of same topic or entity make sure to check by using group for duplicate checks*
- *If you see the cardinality is 1 to 1 relationship between two tables . This is really safe to merge those two tables because Power BI automatically check the values inside those two tables and detect the Cardinality .*

Next Important step – HOUSE Keeping



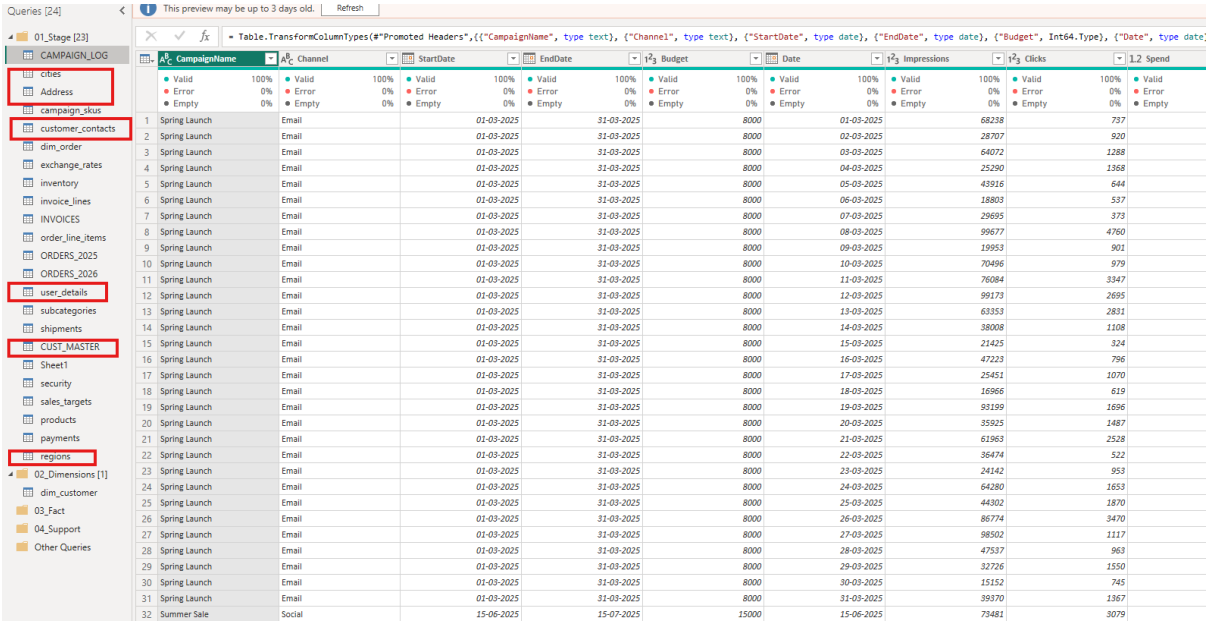
*Once you done with one topic and converted all tables into one**, we have to Clean Up and Organize and following the standards and polish everything in order to make dimensions is ready to get used in the data model.***

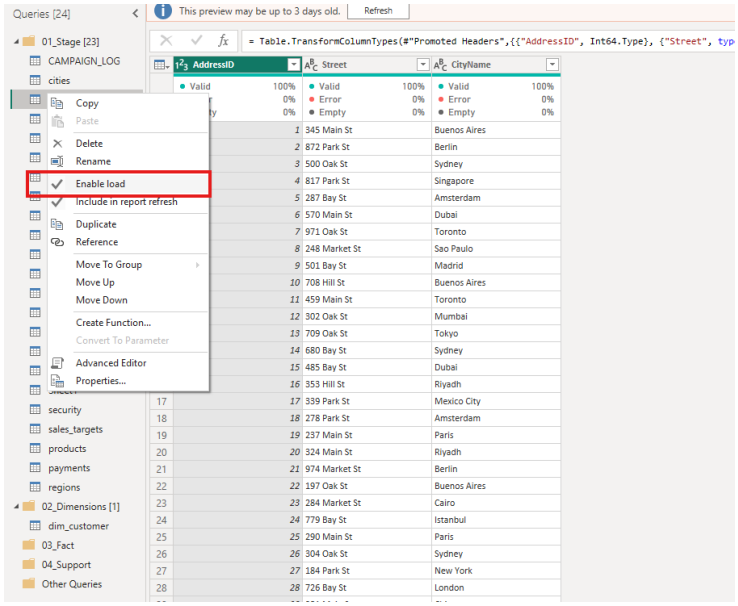
Steps

- First step we have to do is that we have to get rid of unnecessary information in the prepared dimension.. Like unnecessary columns

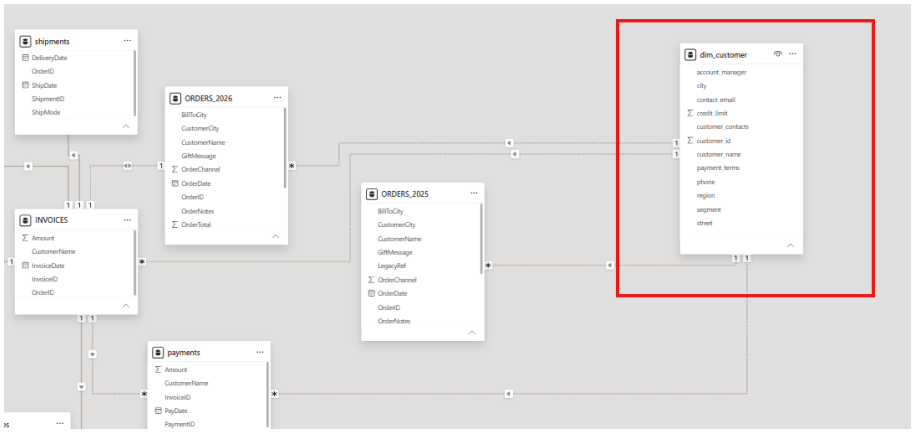
Remember the RULE** - **Every column must earn its place. If its not needed, drop it

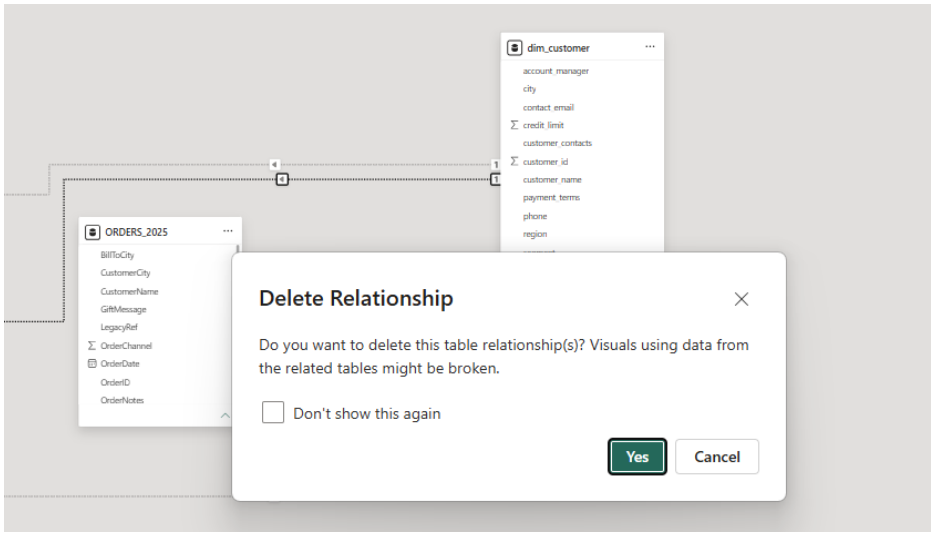
- Once done dropping columns, As per standard we have to keep all the columns exit in the dimension should be in **standard case**
- Important - After all above go and **deactivate all the original tables from the stage one folder** , so that we don't seen them inside the data model anymore.



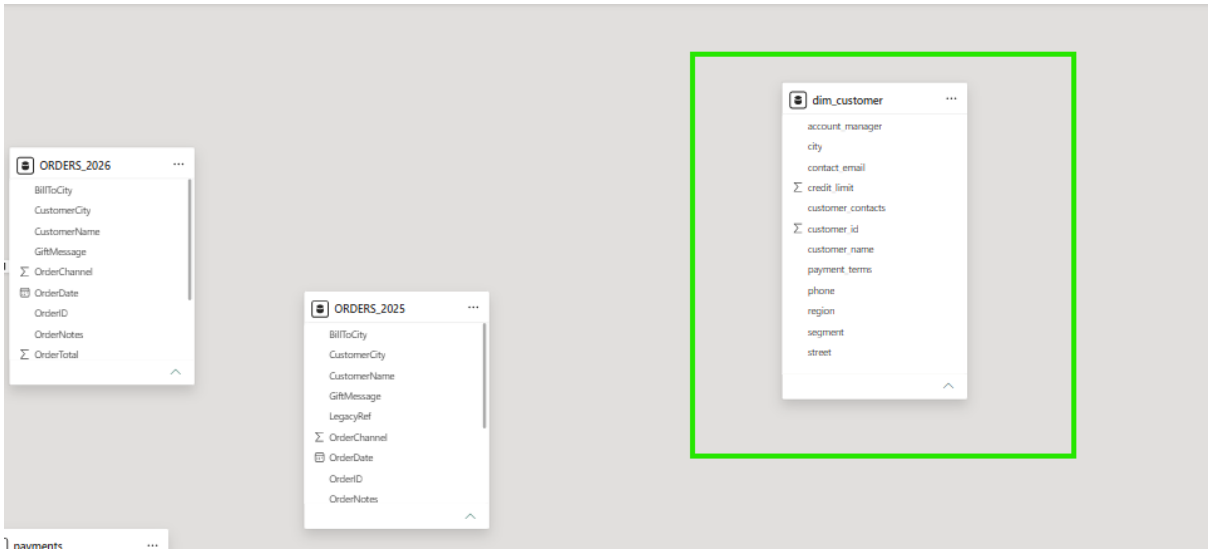


- Once done close and apply and check on data model in BI.
- **Remove** the relationship all other tables which are connected with our ***newly created dim table by BI Automatically***





There we have one good solid **dimension**



Lets start same process again by choosing next topic – Products to prepare a dimension

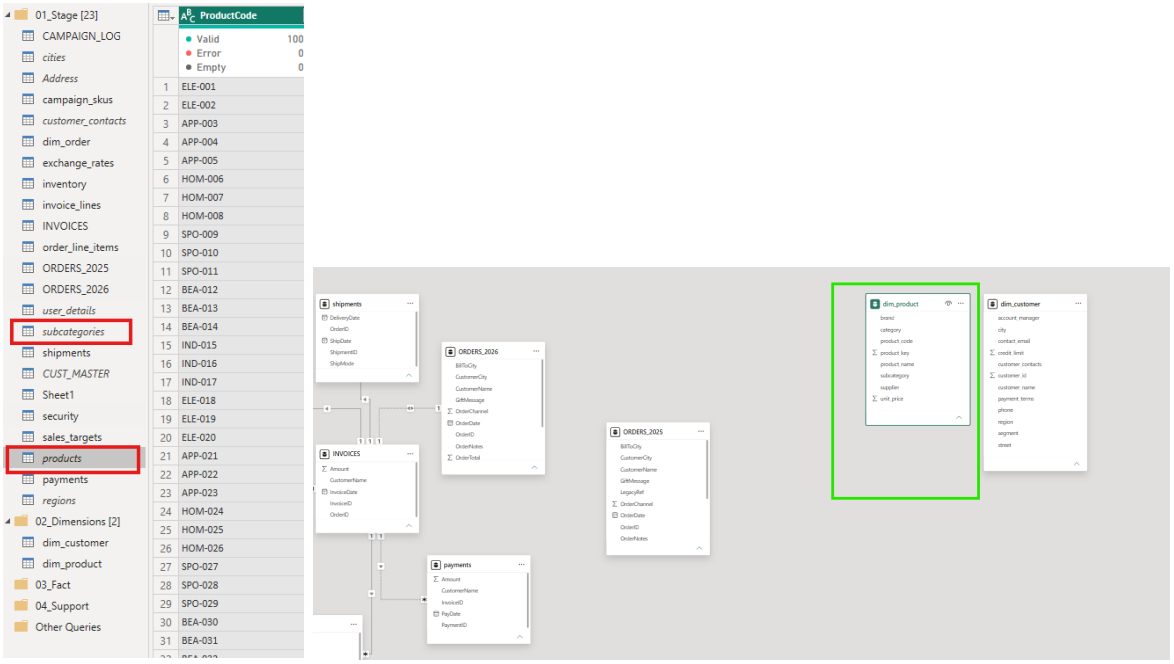
- Note while merging data make sure both tables keys are in same format and style





Dimension 2 – ***same process as per first dimension***

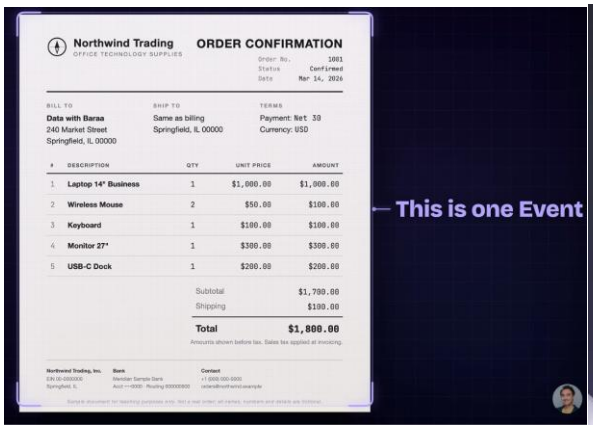
- After all merge , start removing unnecessary columns
- Standard keys, standard capitalization
- Once done all check, go ahead and deactivate the tables from the stage 1 folder in power query
- Once done close and apply and check on data model in BI.
- **Remove** the relationship all other tables which are connected with our ***newly created dim table by BI Automatically***

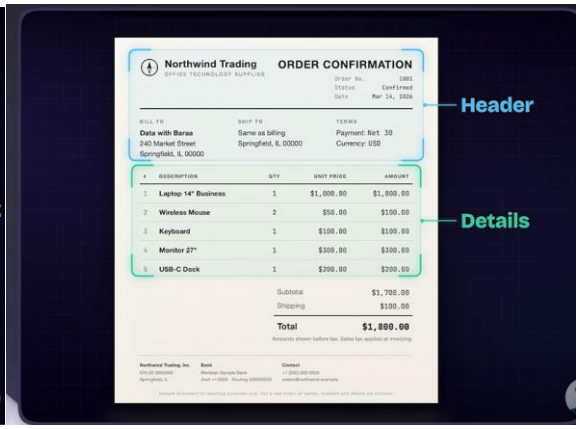


Now we have clean two dimensions, dim_customer and dim_product

Phase 3 – Building Facts

- Choose one event & understand the grain
- Just move and group in data model for refer
- Open power query editor
- Read those tables – ***Grains*** what one row represents
- ***This is the complicated process in data modeling , so follow the process clearly***

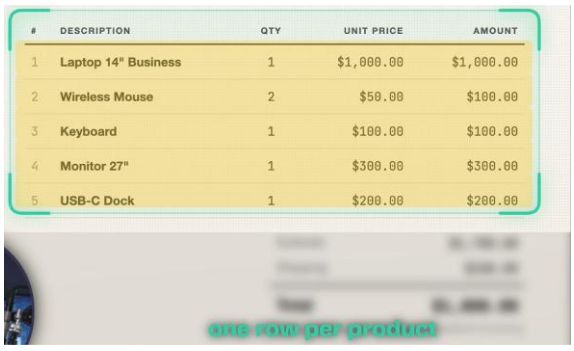




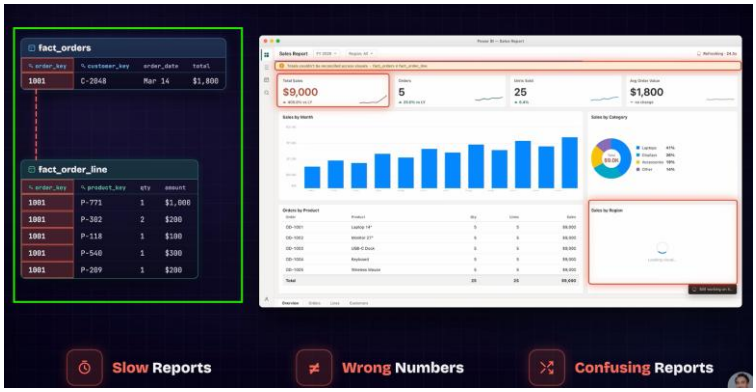


- Order Confirmation

Header **(TOP LEVEL)** , we have those information's only once for each order right . Those are header, **Details** **(Lower Level)**

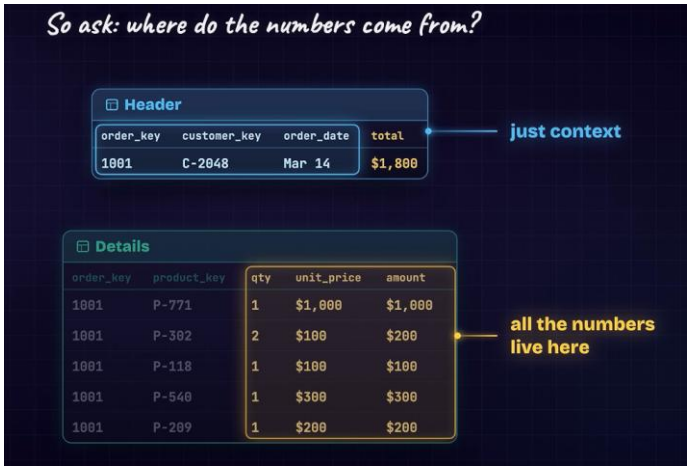


Mistakes – Many will see and consider ok we have 1 vs * so we can join, but it wont give correct report



Rule for fact Tables





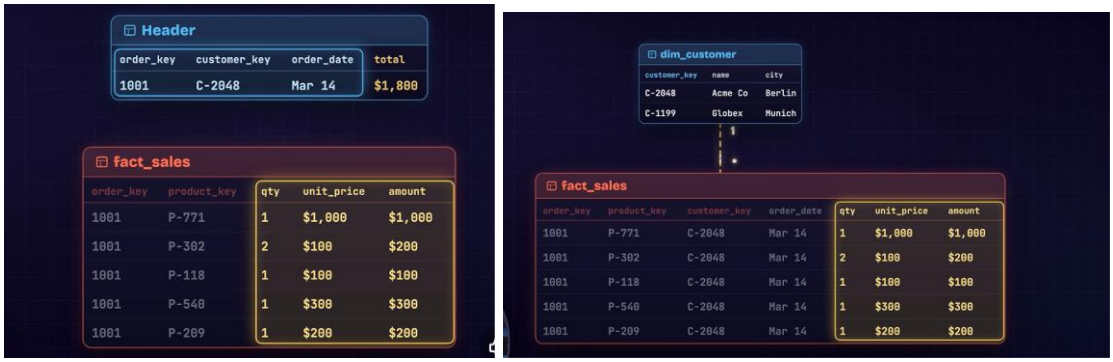
- If you see all the numbers are actually stored and comes from details .
- Header is just like giving us some context and may be like max one or two number, but the core numbers , measure are in the details.
- **So that means it is always better to build a fact table that comes from finest details from the lower level,** so that means we can always roll up and aggregate the data back to the header. In most cases we going to create core fact table from the details and we are going to the header in order to get more context for our new facts.

- So either we are going to use it in order to generate a new dimension form the header or we gonna use as well to connect the data model to our new fact from the details.

Final Summary

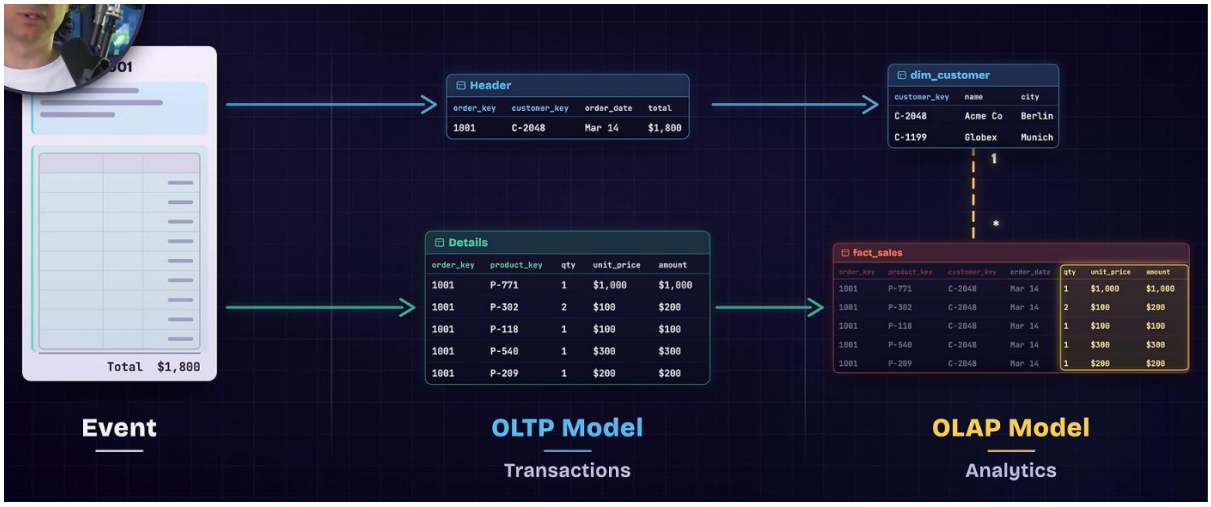
I repeat - **It is always better to build a fact table that comes from finest details from the lower level,**

Convert the both tables **Header and Fact sales(OLTP)** into one fact table at lower level (second image) . Means we gonna to create the core fact table from the details and we gonna header in order to get more context for our new facts. In Order we can generate new dimensions from the header or we going to use it as well to connect the data model to our new fact table from the details.



Always follow this pattern

This is very common pattern, we make fact from the details and we use the help of the headers in order to generate dimensions and to connected to our fact.



These are the base foundations for fact tables.

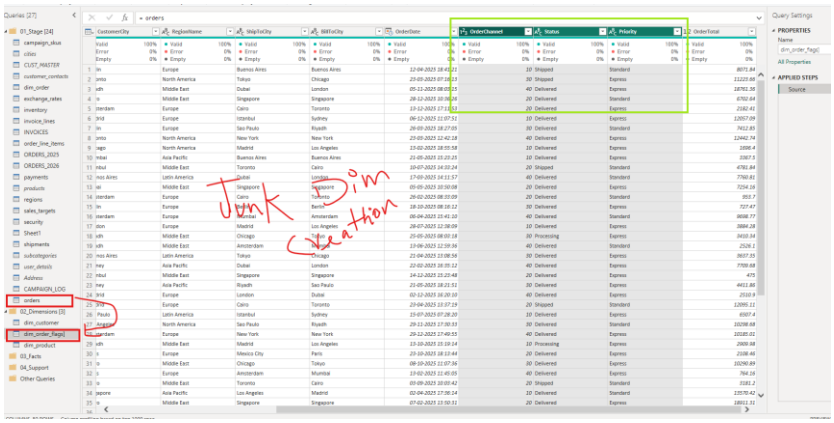
Preparation of Fact table

- Go to the table in PQ, lets combine tables.
- If we have two tables same grain , they are identical and have same shape , we can go and append them.
- Now we have two tables with same grain and identical. So lets append
- Append Orders 25 and 26 and once done, remove unwanted columns for reporting.

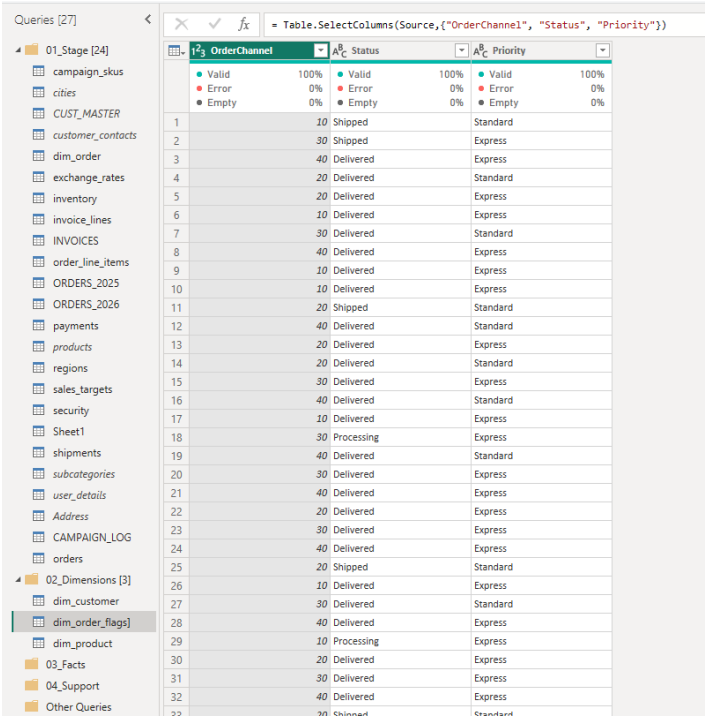
Once done review the appended table and see any context columns are ready to create a new dimensions that fact tables if not available already

Creating dimensions from fact table

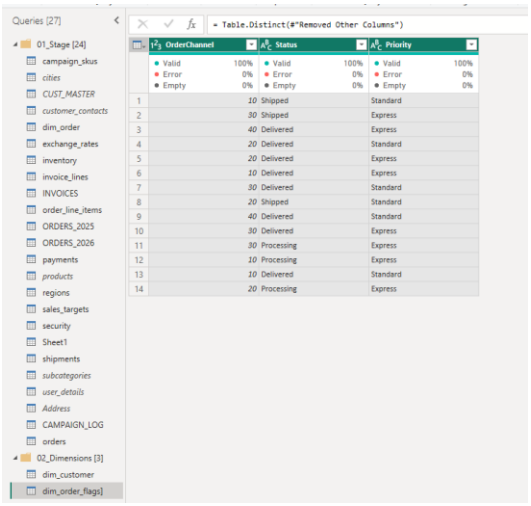
Step 1 – Review the table and choose columns that are eligible for dim



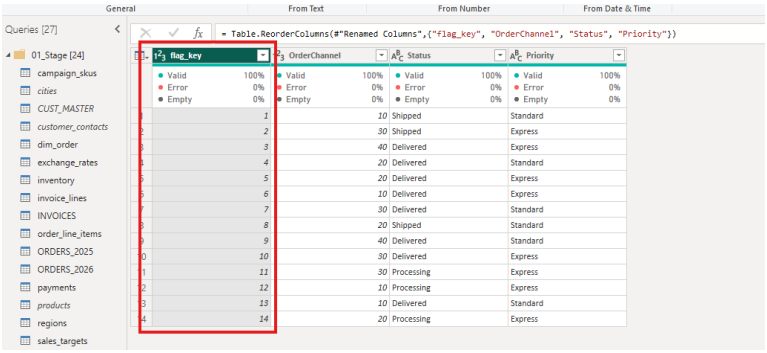
Step 2 – Remove all other columns

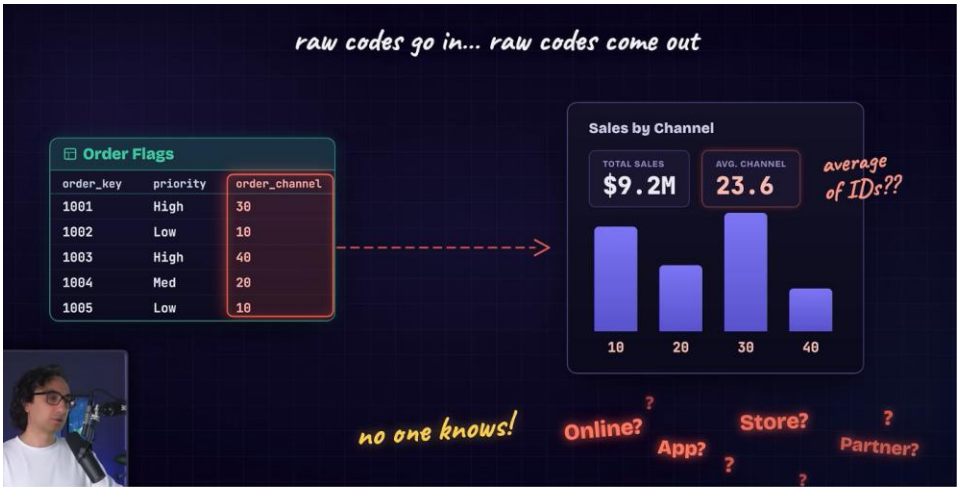


Step 3 - Remove duplicates

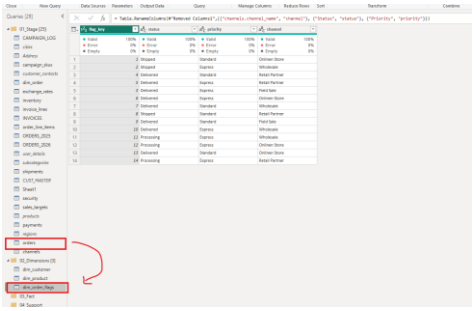


Step 4 – We need a unique Key in dimensions to connect with facts later



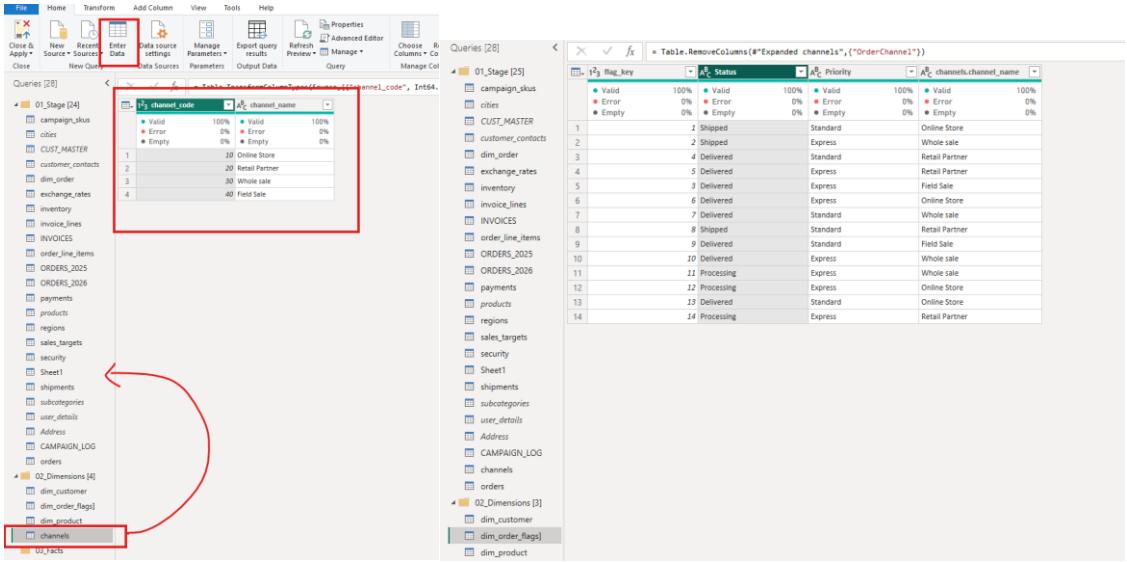


Date enhancement



Create New Table and assign channel code accordingly

- Move the created table to stage and merge with the required table ***(order_flag)***



Before



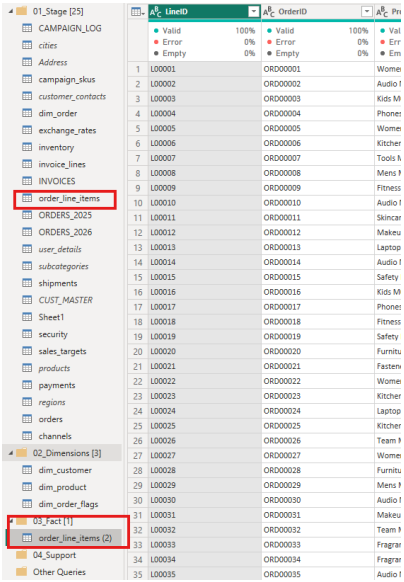
After adding channel names – so we need to add something frm ur side for better report



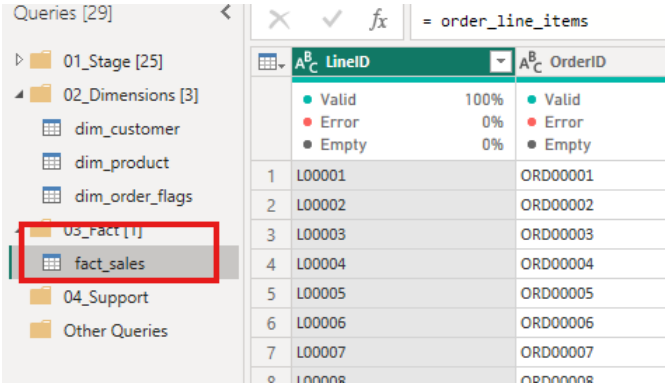
- *Once done, make sure to polish our table*

Now lets go to lower level fact table – **Orders lines**

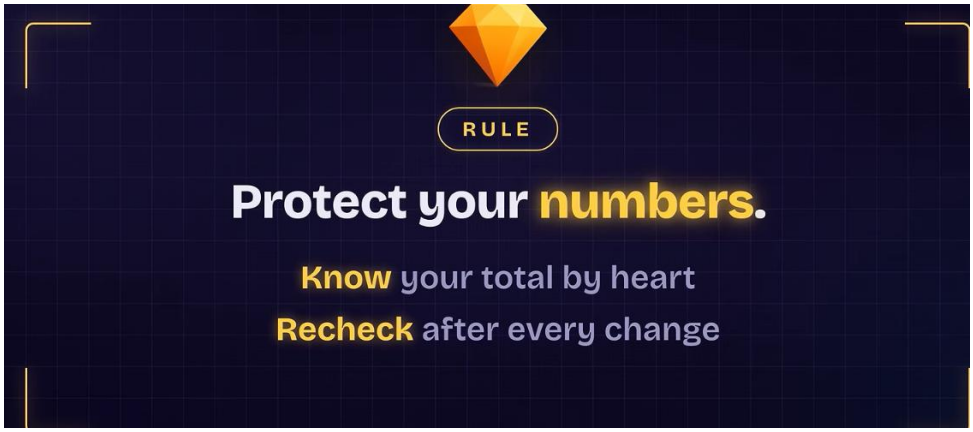
- Make reference of orders lines items



Give a name

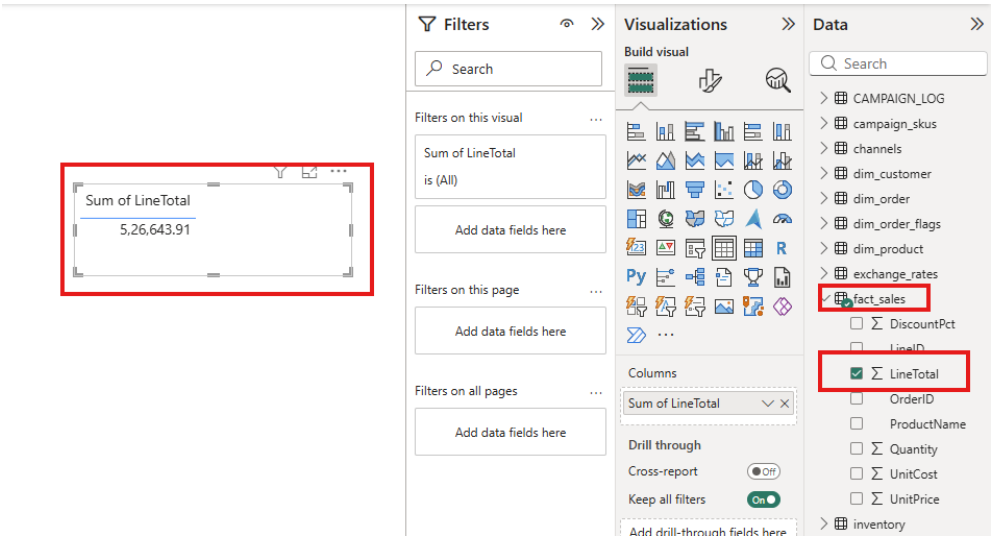


After adding the lower level details ordes line item to fact table , now we need the dimensions , so we have already here the product name, but we need the customers , junk dimensions , we need order date , so we need all those stuff from the header to be included as well in our new facts. So we need to start combining to out new fact and here you have to be very careful, if you do one wrong merge , whole fact will get collapse

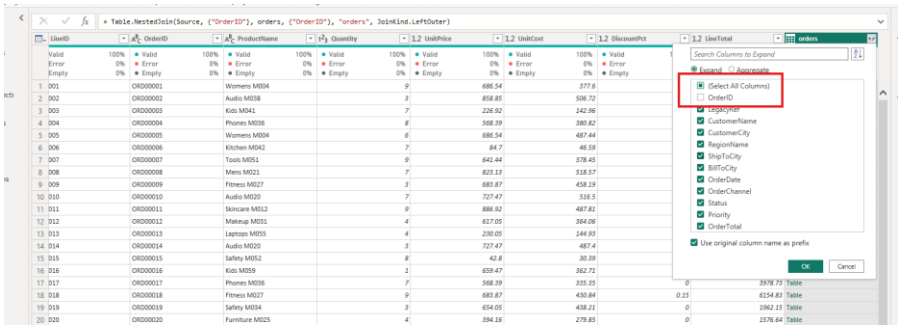


How to do this & keep tracking table is not affected by numbers

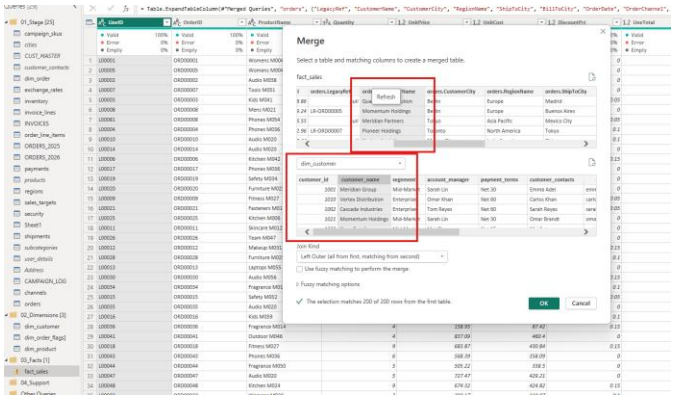
- Go to report view
- Pick **one important measure** from the **fact table** and see the number or value and just keep in mind or note some where
- That is the number we have to protect it, any thing we combine we add to this fact table . ***It always should result exactly to this same number, if anything changed we have an issue.***



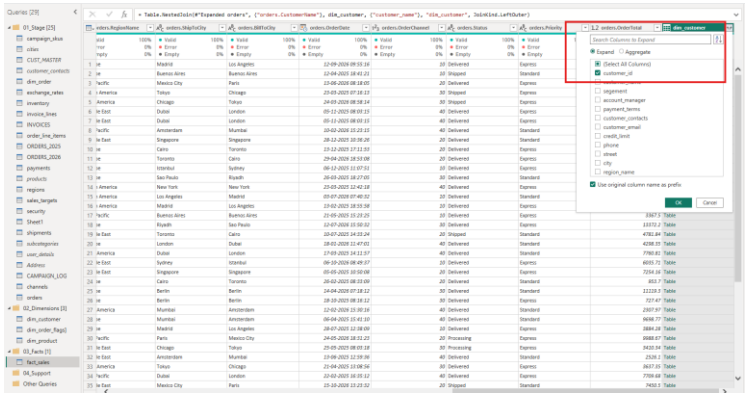
- Now start merging the fact tables Lower level(Order line items) with **header** - top level table (Orders) – Merge the table as like this



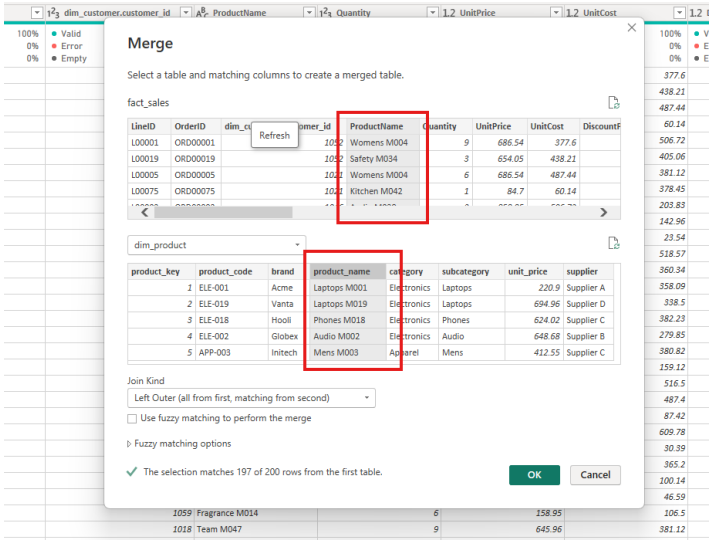
- Go and check any changes in number in report view
- Next step we can go and build keys inside the facts. You know each fact has lot of keys in order to connect things to the dimensions. And our fact is not ready yet.
- Lets say in fact table instead of having product names we can go and have the product key, same thing for the customers. So lets do that. Bring those ids to the fact. This is called lookup
- But this time we are merging with the dimensions like below
- Our target is to get cx ID from dimensions to fact tabel



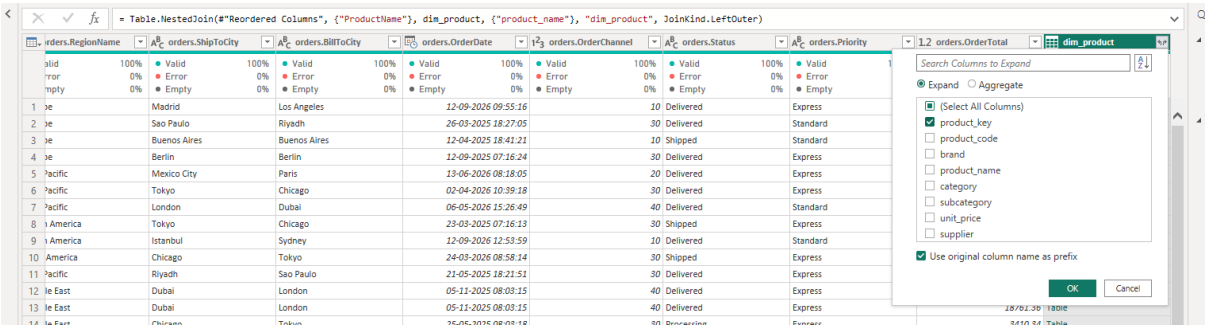
We just want one thing from the dimension that is customer ID so only choose that column to merge.



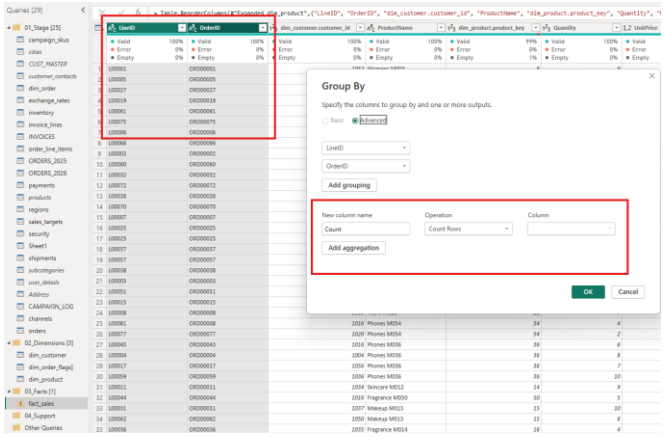
- Once done, Go and check any changes in number in report view
- Lets do same process with another dim product table key to fact

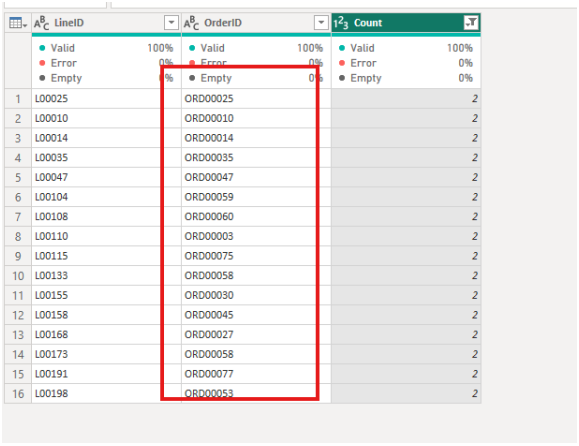


- Our target is to get Product ID from dimensions to fact table and align in the table



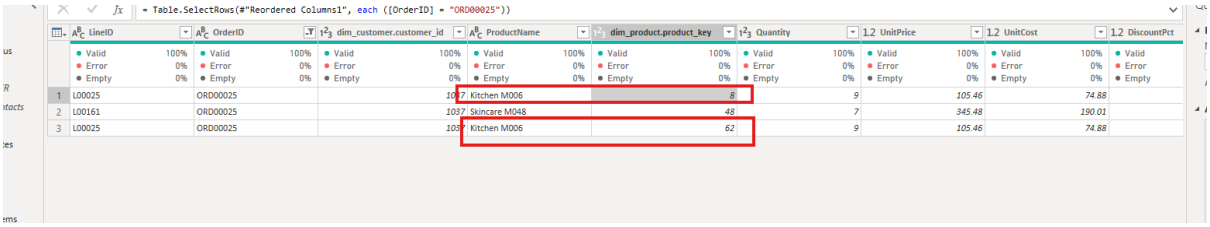
- Once done, Go and check any changes in number in report view
- **There is a change in numbers , *now review the fact sales which order has issue***





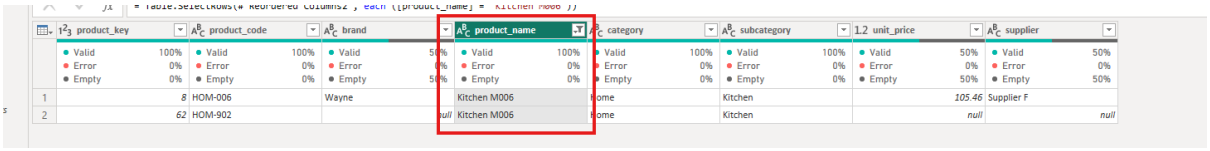
	LineID	OrderID	Count
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	L00025	ORD00025	2
2	L00010	ORD00010	2
3	L00014	ORD00014	2
4	L00035	ORD00035	2
5	L00047	ORD00047	2
6	L00104	ORD00059	2
7	L00108	ORD00060	2
8	L00110	ORD00003	2
9	L00115	ORD00075	2
10	L00133	ORD00058	2
11	L00155	ORD00030	2
12	L00158	ORD00045	2
13	L00168	ORD00027	2
14	L00173	ORD00058	2
15	L00191	ORD00077	2
16	L00198	ORD00053	2

- Go and search those duplicates individualy , why two entry



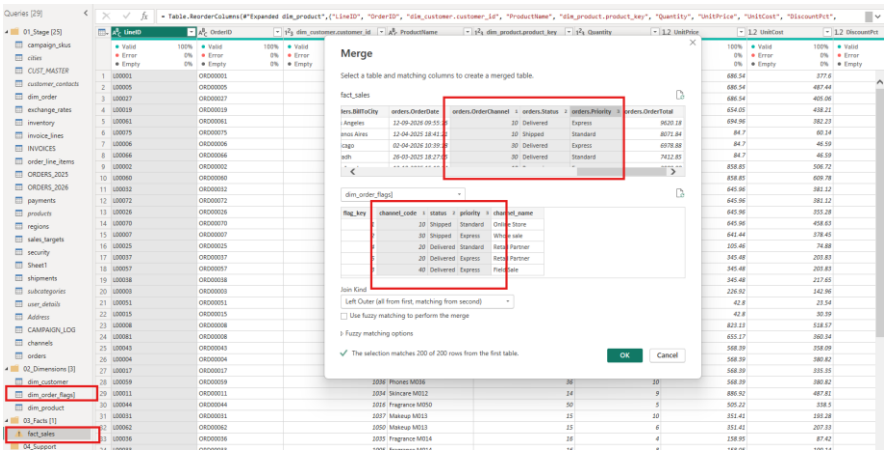
	LineID	OrderID	dim_customer.customer_id	ProductName	dim_product.product_key	Quantity	UnitPrice	UnitCost	DiscountPct
1	L00025	ORD00025	1027	Kitchen M006	8	9	105.46	74.88	
2	L00161	ORD00025	1037	Skincare M048	48	7	345.48	190.01	
3	L00025	ORD00025	1037	Kitchen M006	62	9	105.46	74.88	

- Now it seems there is issue in product dimensions, so go and check there

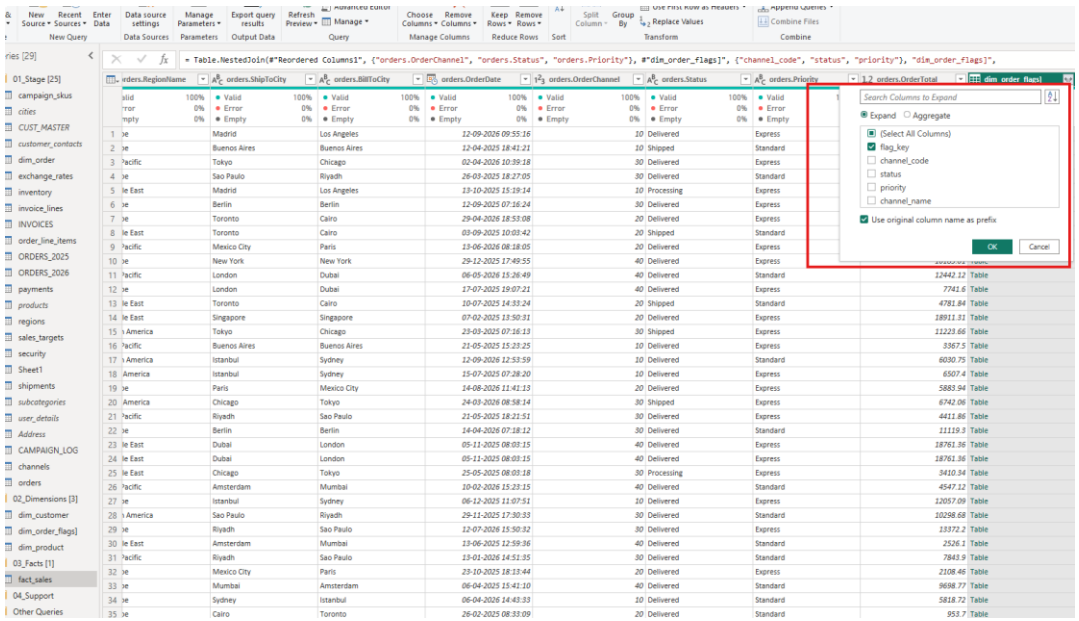


	product_key	product_code	brand	product_name	category	subcategory	unit_price	supplier
1	8	HOM-006	Wayne	Kitchen M006	Home	Kitchen	105.46	Supplier F
2	62	HOM-902	null	Kitchen M006	Home	Kitchen	null	null

- We found there is duplicate inside dimension, so it is data quality issue we go from source file itself , so we can go and removing this and as well inform team to remove
- For now we can do it manually in PQ on source file itself in Step, by check null or something else on your convince.
- Once done now check the report view whether the numbers are correct.
- Check which dim are pending to add in fact table , now order flags



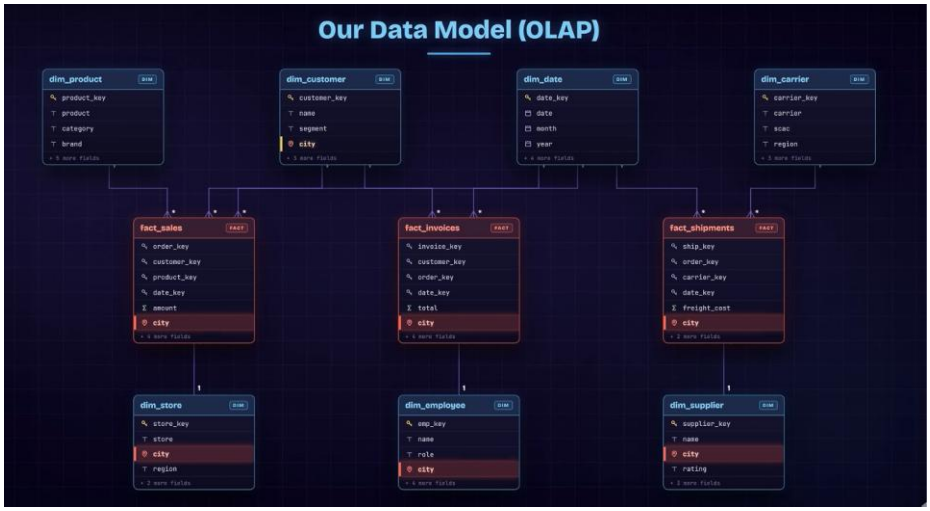
As usual keep only key

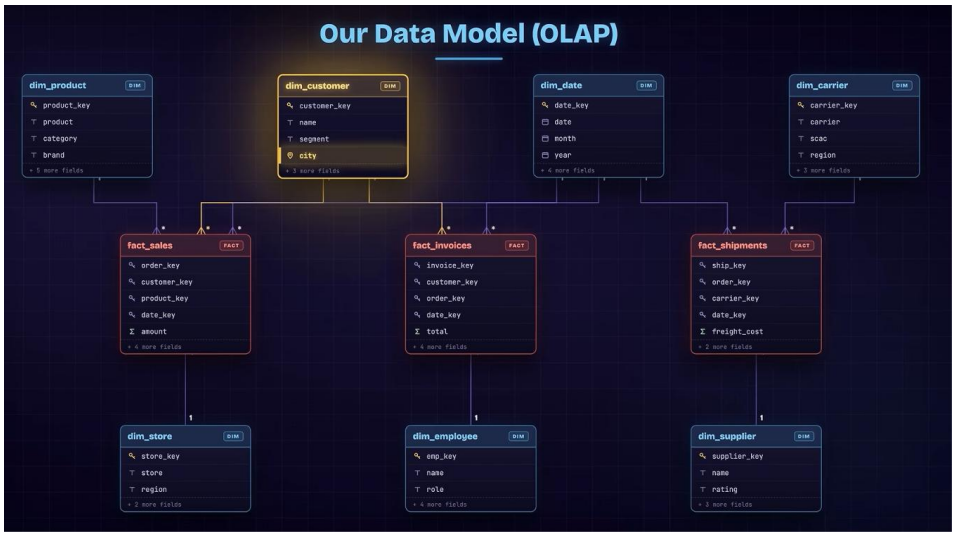


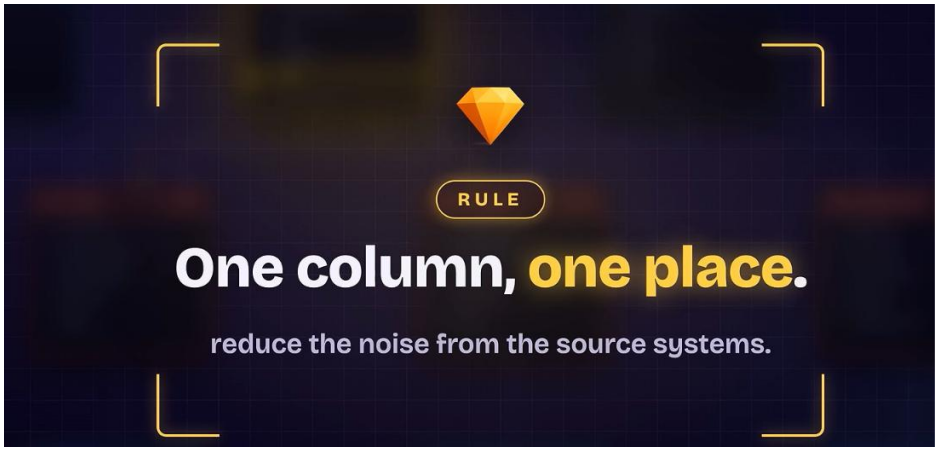
- Once done, Go and check any changes in number in report view
- So now we have we got all the keys id are inside our fact and it is ready to get connect with dimensions
- Now go to next phase go and drop things and polish the facts in order to follow the standard and the rules we have added at the beginning

Information - In transactional source system they usually store the same information over and over In different table in order to optimize the performance of their UI app. This is big issue for us when we have like same information every where inside our data model, this is going to make a big confusion on how to build a report's .





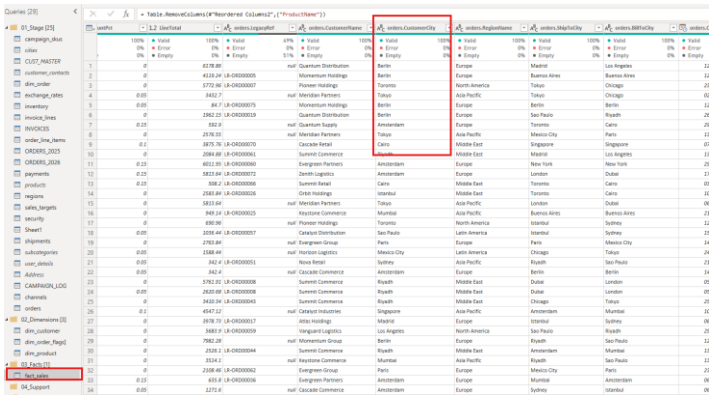




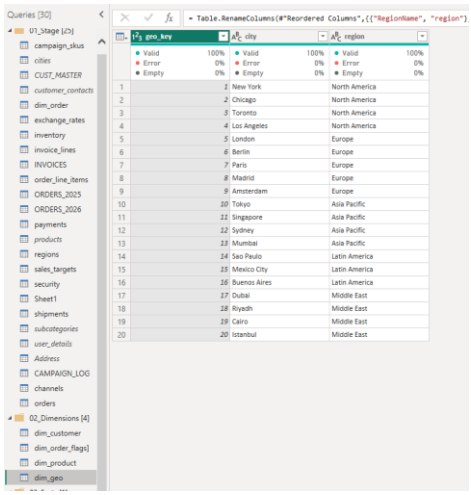
So we have to protect the data model by having one single point of truth for each info. So that means if I have customer city inside the dimension customer. I don't need it somewhere else . this gonna be the dim customer is the single point of truth for the cities of the customers. So we have don't have to leave it in another dimension or to leave it in the facts. So always try to reduce the noise .

Example :

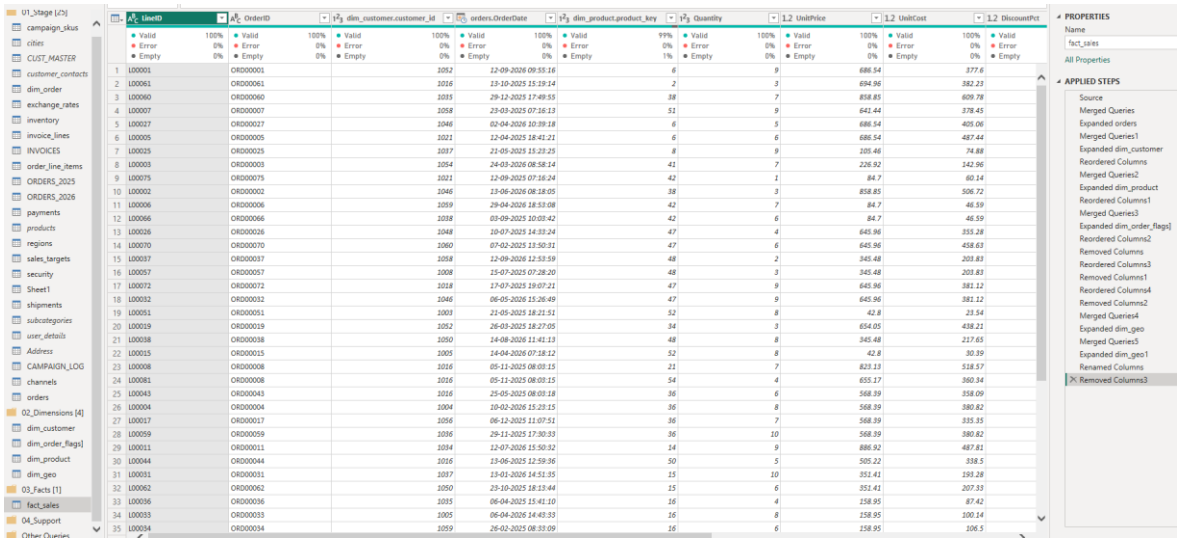
- we have customer city in Fact table and the same is in dimensions , so just go and remove it from fact



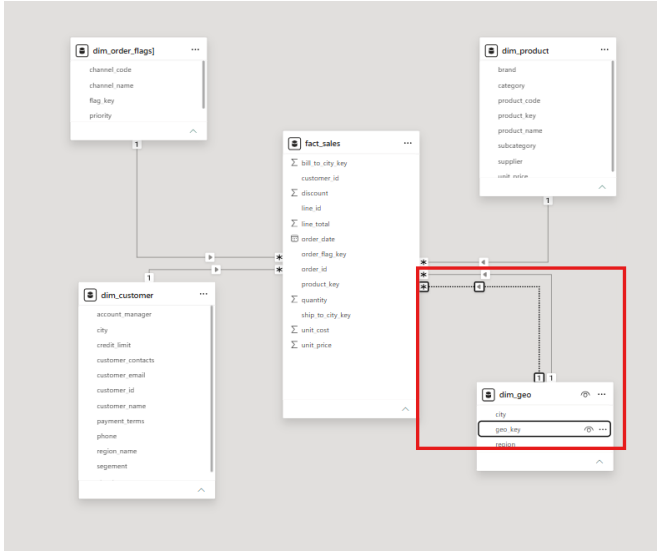
- Same like region column
- Also remove any pre aggregated value in fact , example Order Total
- We can prepare ship city and bill city as dimensions and merge with keys and remove all other and keep only keys in fact



- Now look at our fact we have only numbers and dates. And this is exactly how the fact table supposed to be.

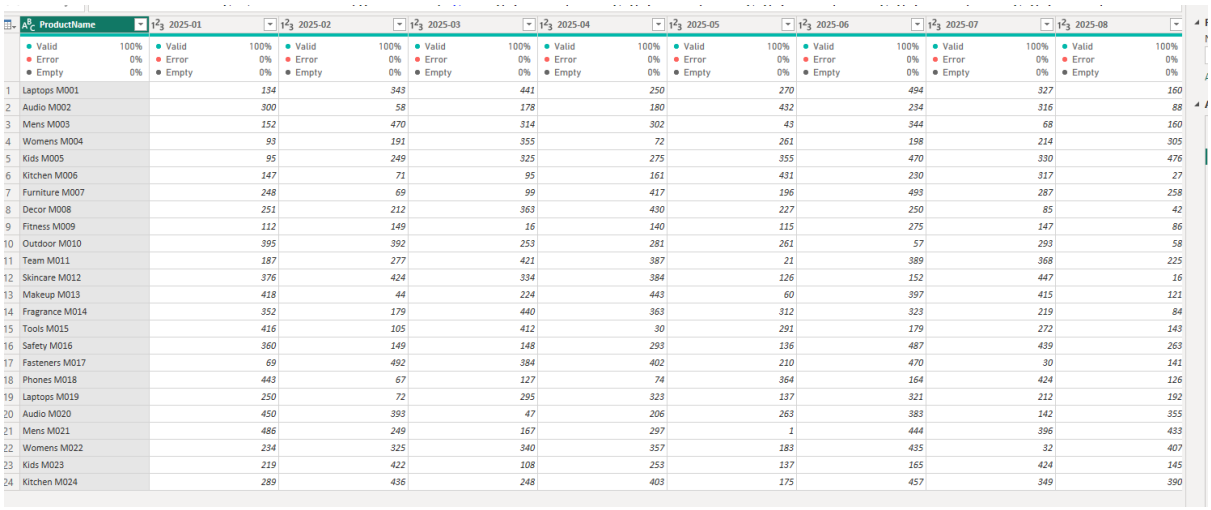


Now go and deactivate tables in stage one and we can connect with dimensions to our fact.



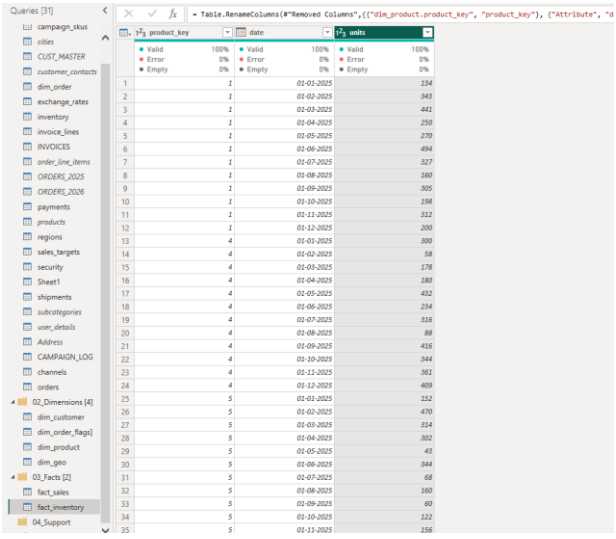
Building Inventory Fact

- Same process – Make a reference and move to fact folder and proceed



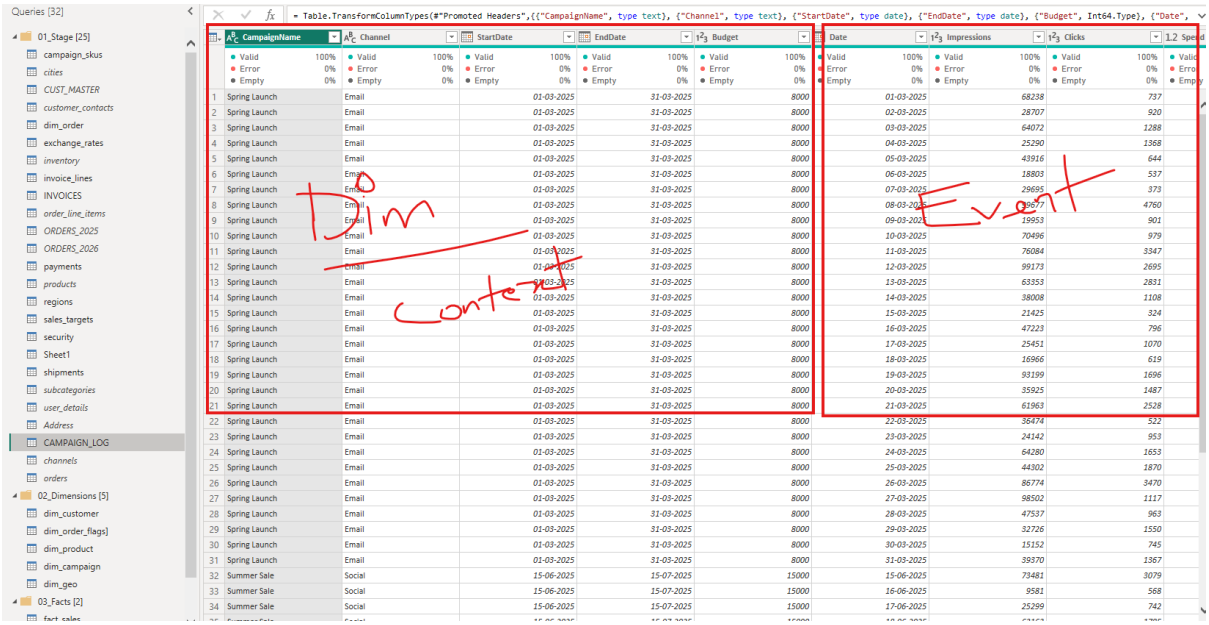
	ProductName	2025-01	2025-02	2025-03	2025-04	2025-05	2025-06	2025-07	2025-08
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	Laptops M001	134	343	441	250	270	494	327	160
2	Audio M002	300	58	178	180	432	234	316	88
3	Mens M003	152	470	314	302	43	344	68	160
4	Womens M004	93	191	355	72	261	198	214	305
5	Kids M005	95	249	325	275	355	470	330	476
6	Kitchen M006	147	71	95	161	431	230	317	27
7	Furniture M007	248	69	99	417	196	493	287	258
8	Decor M008	251	212	363	430	227	250	85	42
9	Fitness M009	112	149	16	140	115	275	147	86
10	Outdoor M010	395	392	253	281	261	57	293	58
11	Team M011	187	277	421	387	21	389	368	225
12	Skincare M012	376	424	334	384	126	152	447	16
13	Makeup M013	418	44	224	443	60	397	415	121
14	Fragrance M014	352	179	440	363	312	323	219	84
15	Tools M015	416	105	412	30	291	179	272	143
16	Safety M016	360	149	148	293	136	487	439	263
17	Fasteners M017	69	492	384	402	210	470	30	141
18	Phones M018	443	67	127	74	364	164	424	126
19	Laptops M019	250	72	295	323	137	321	212	192
20	Audio M020	450	393	47	206	263	383	142	355
21	Mens M021	486	249	167	297	1	444	396	433
22	Womens M022	234	325	340	357	183	435	32	407
23	Kids M023	219	422	108	253	137	165	424	145
24	Kitchen M024	289	436	248	403	175	457	349	390

Here data to stored as human understanding view we need to convert it to data model report and analysis view, from to pivot to unpivot

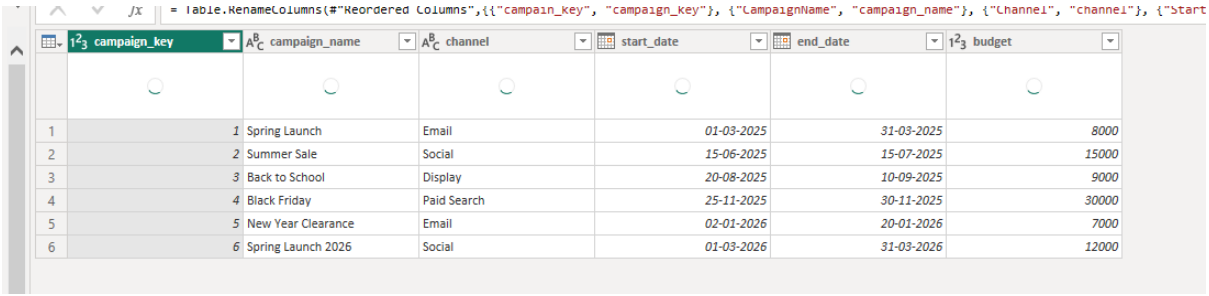


Once done go and deactivate, on stage one inventory file

Building fact & dimension campaign

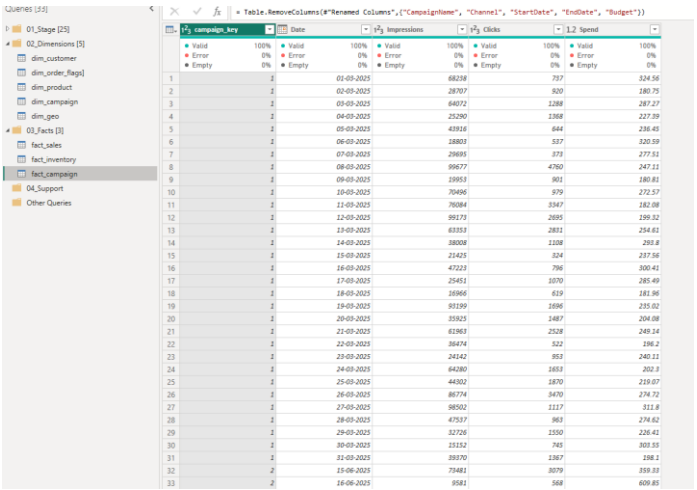


Dim_campaign

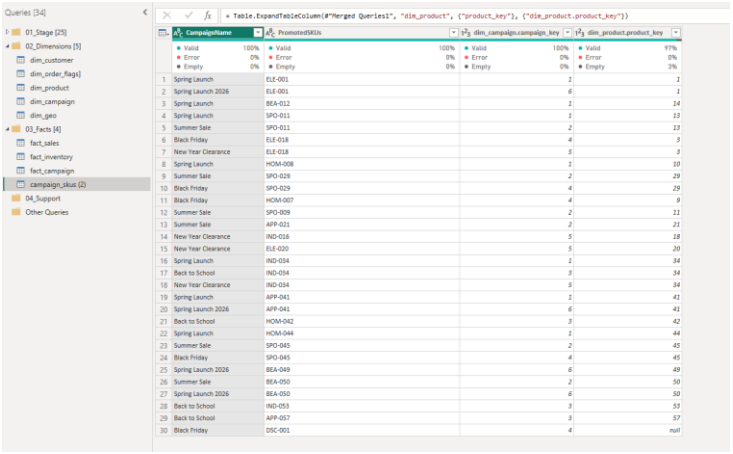


	campaign_key	campaign_name	channel	start_date	end_date	budget
1	1	Spring Launch	Email	01-03-2025	31-03-2025	8000
2	2	Summer Sale	Social	15-06-2025	15-07-2025	15000
3	3	Back to School	Display	20-08-2025	10-09-2025	9000
4	4	Black Friday	Paid Search	25-11-2025	30-11-2025	30000
5	5	New Year Clearance	Email	02-01-2026	20-01-2026	7000
6	6	Spring Launch 2026	Social	01-03-2026	31-03-2026	12000

Now lets go and create fact table and merge dim campaign key with fact_campain and remove the unwanted columns which are already in dim

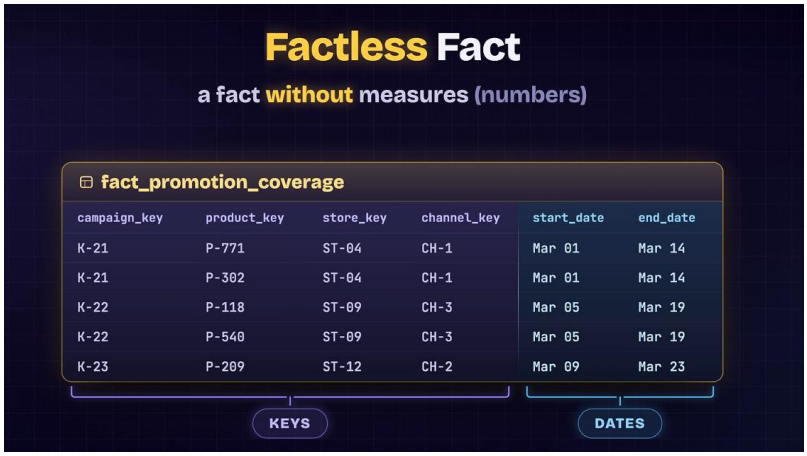


Campaigns skus

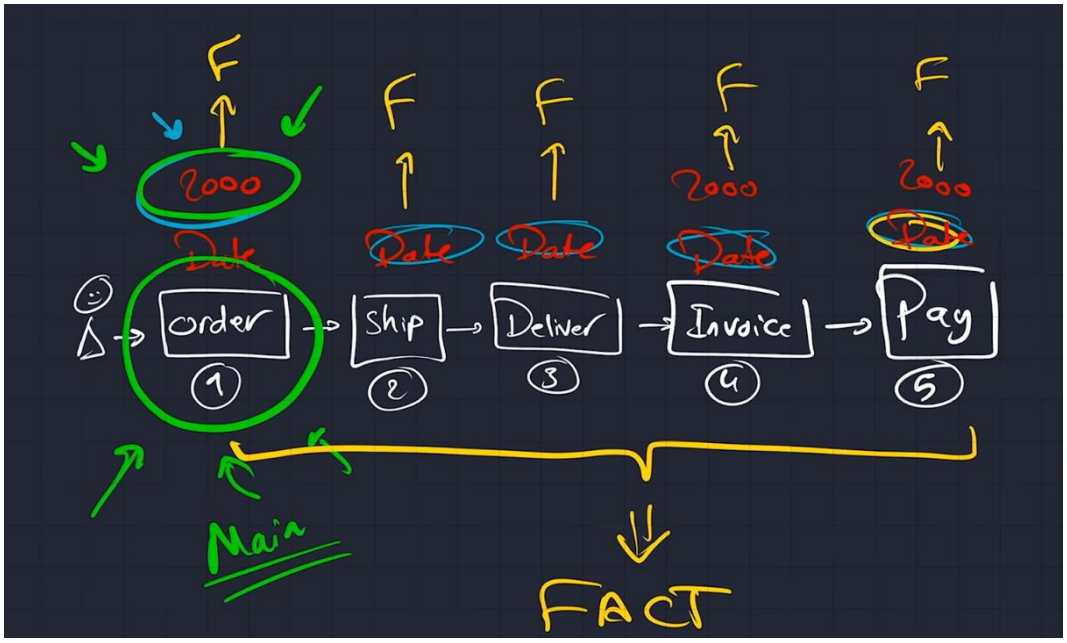


This is call factless fact

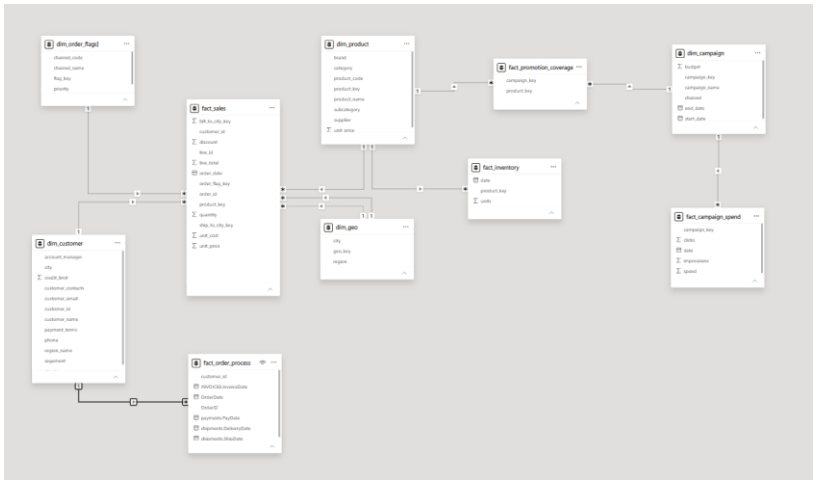




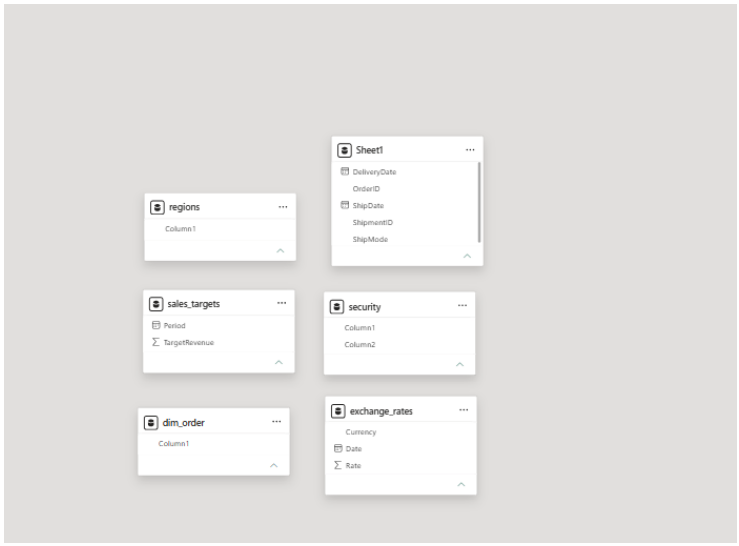
Building the Order fullfilment fact





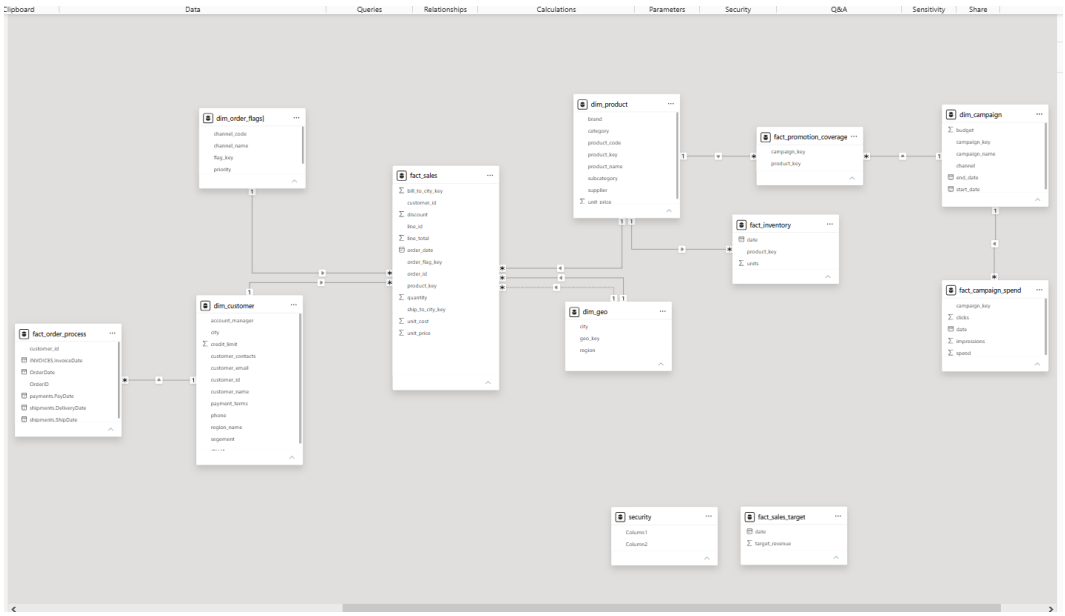


Balance tables –



Lets review and delete or deactivate unwanted tables.

Here we go with healthy **Star Schema** – Data model



Protect the core

- Understand the **business & domain**
- Build only **core facts & dimensions**
- A **separate model** for special cases
- Protect the **core data model**

WATCH OUT

Without saying No,
adding unnecessary tables,
you risk breaking your model.

- wasted space
- slower performance
- confused reports

Phase 4 – Polishing the Model

Step 1

Go and check



- Make sure to check and change Date formats in all tables as YY-MM-DD

Step 2

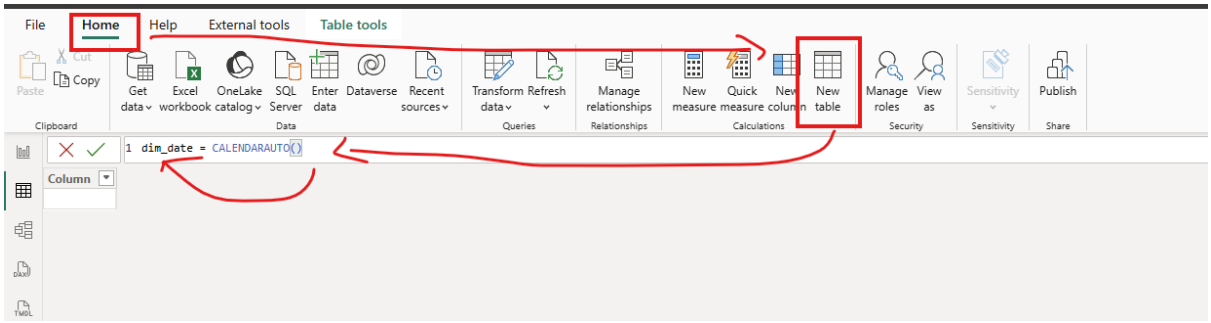
Create a dim_date. It is a special dimension that actually going to be needed almost in all our facts.

So better create it and connect with the data model.



How to create date dimension in data model.

Go to home -> Calculations -> New Table

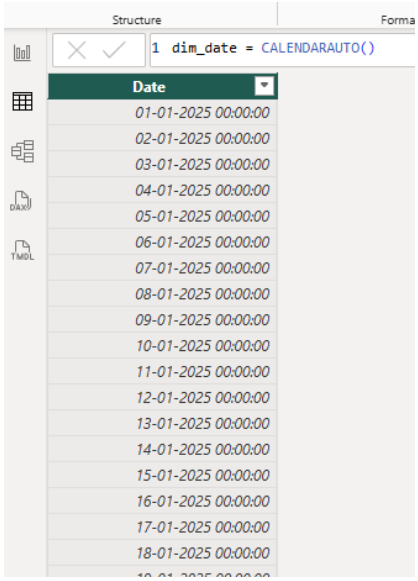


Formula to create

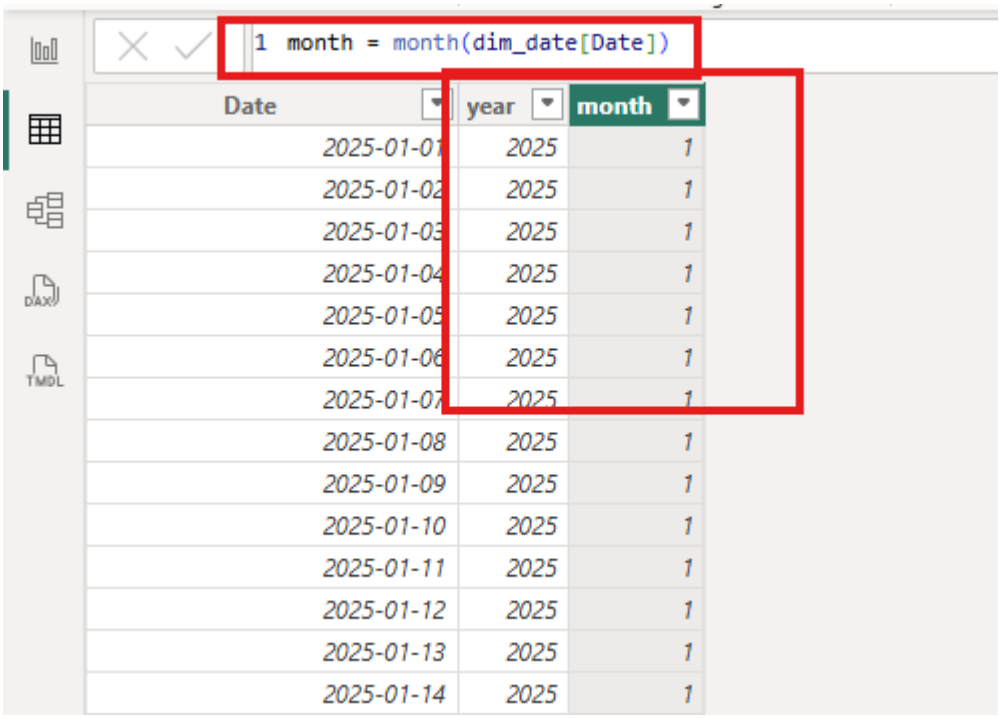
= CalendarAuto ()

This is very cool function in powerBI. So this read and returns a table with one column of dates calculated from the model automatically.

It exactly go and scan the whole data model understand the range of our business. What is the earliest date and what is latest date.

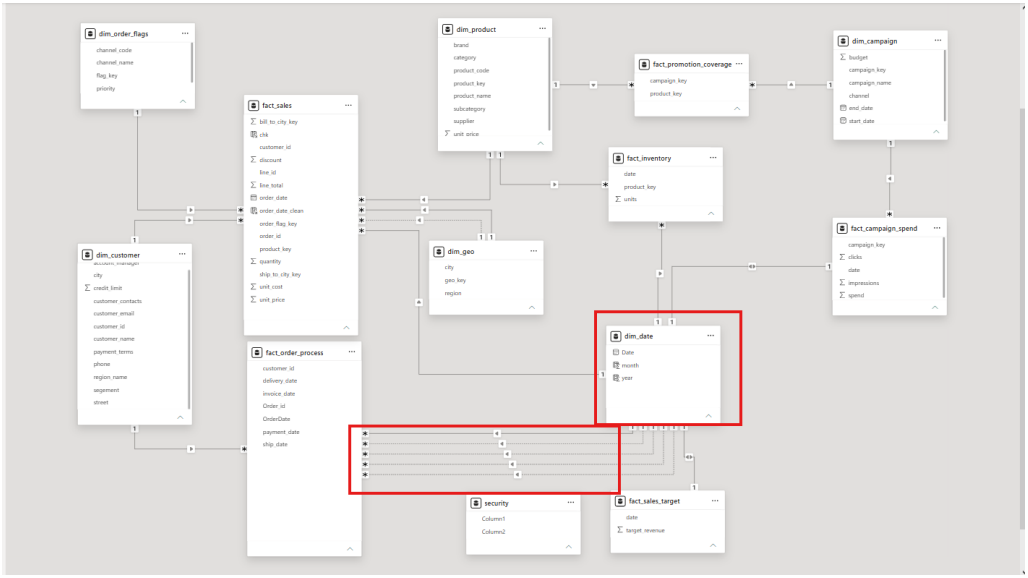


It expands automatically when business grow and when we have 2027 we don’t have to deal with the dimension date cuz it can grow up automatically.



We can also add year , month separately using formula

Lets go and connect dim_date to facts



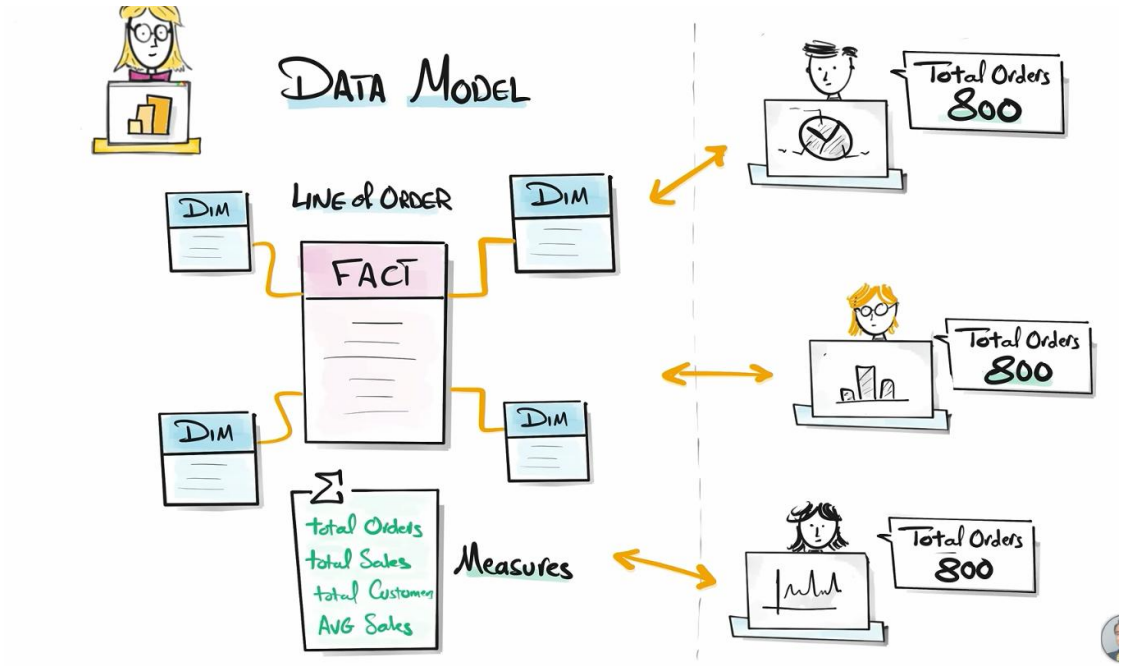
Those many relationships are role playing

Step 3

Build and collection of measures

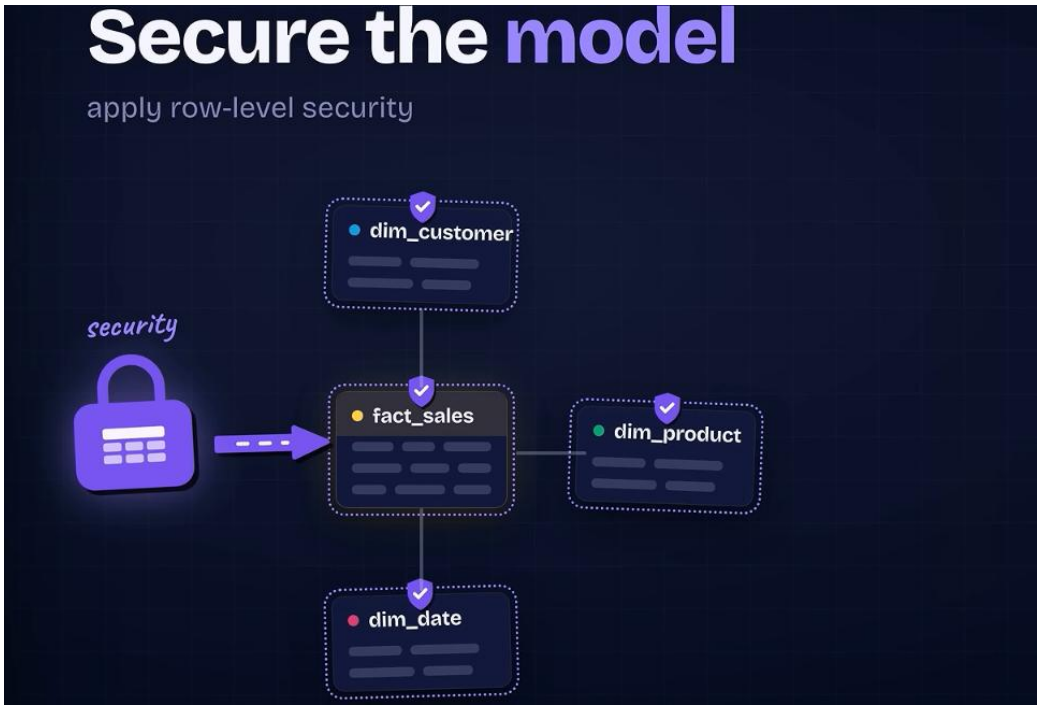


We gonna to make like set of very obvious measure that's going to be needed on top of this model.

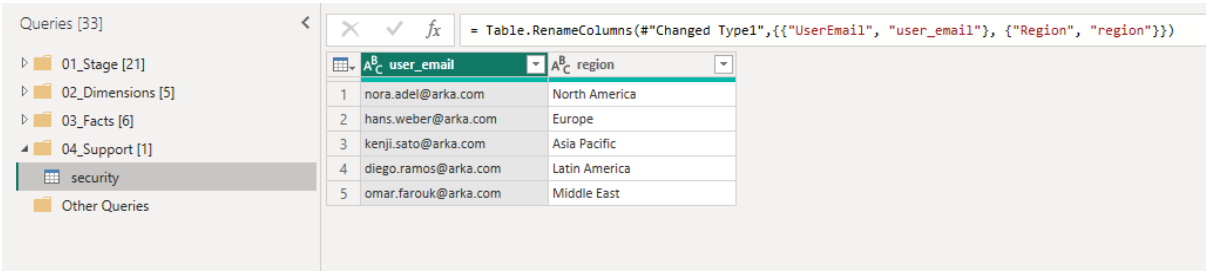


Step 4

*Make sure our data model is secure****(Row level security****)*



Grap the list to other support



Security Requirements

Row-Level Access · Sales data

Sales data is restricted by region. Each user may view only the sales of the region the security office assigns to them.

THE REQUIREMENT

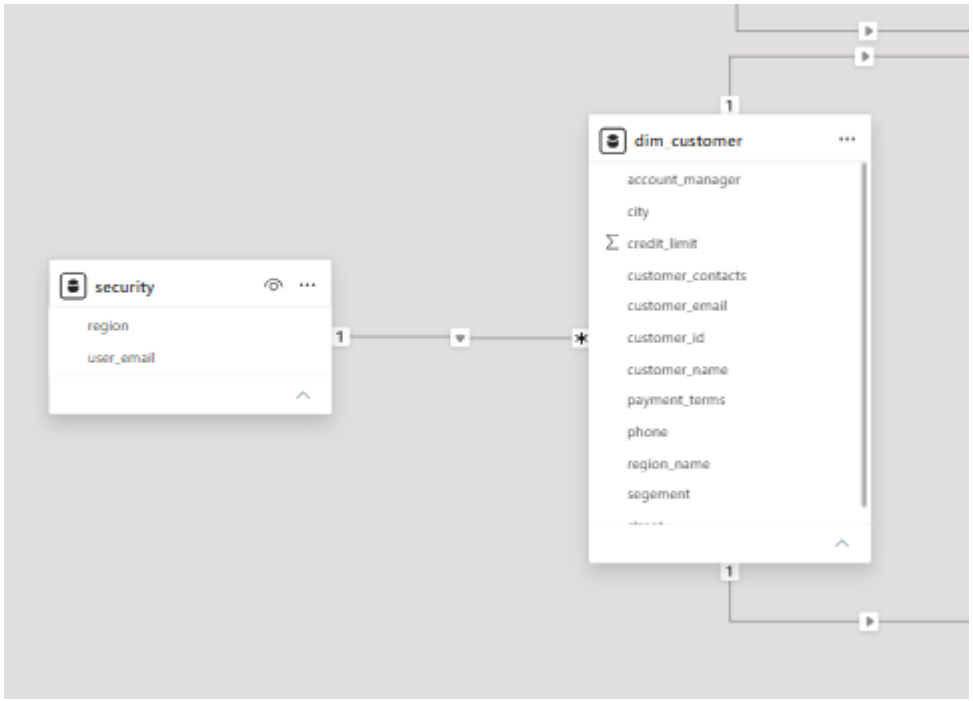
Every user sees only their own region.

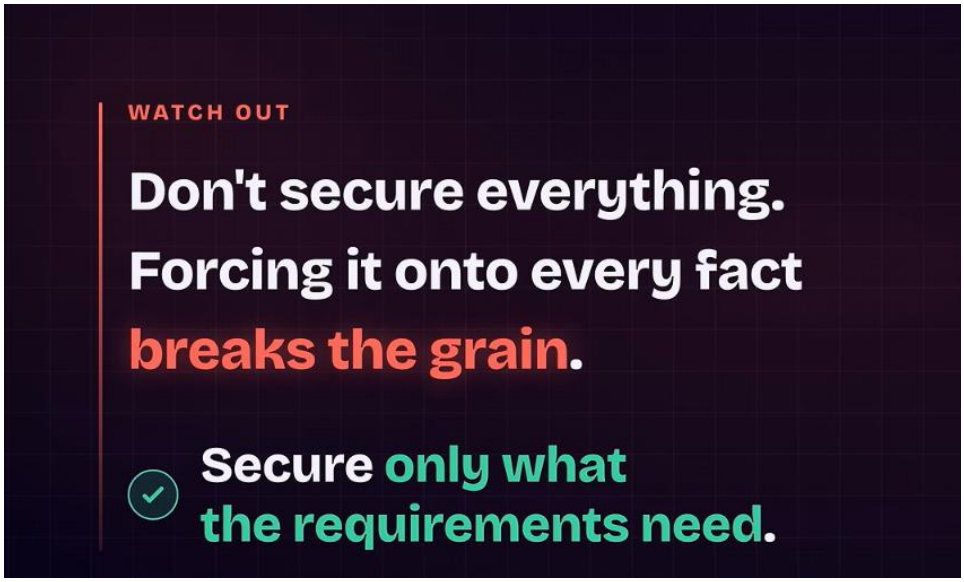
ASSIGNED ACCESS

Nora	North America
Hans	Europe

Security Office
APPROVED BY

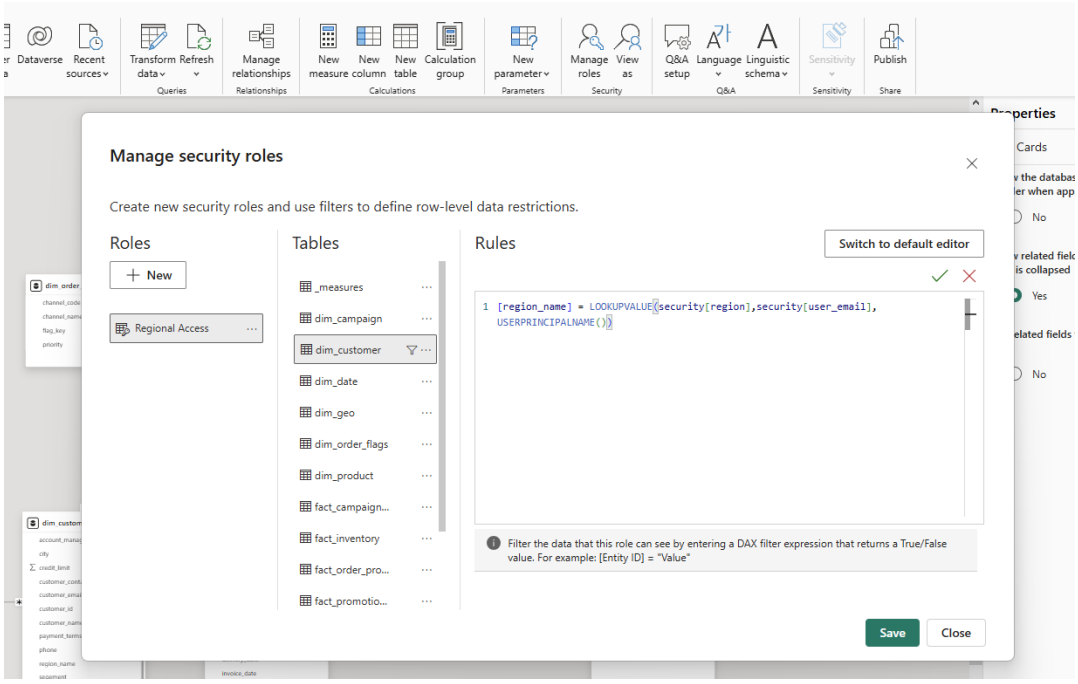
★ DATA SECURITY OFFICE ★
APPROVED
region access · 2026

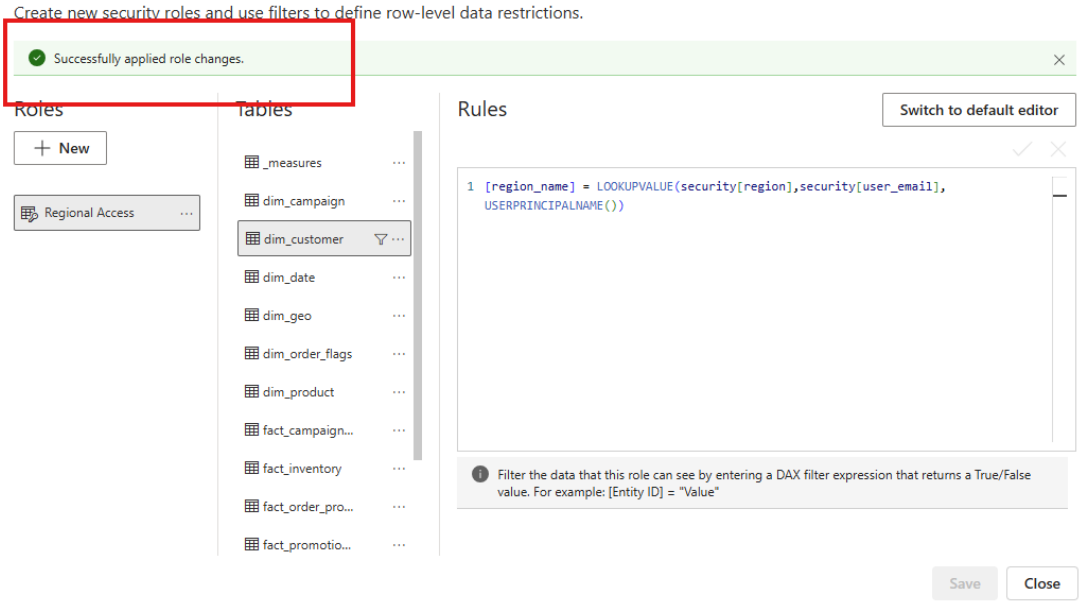


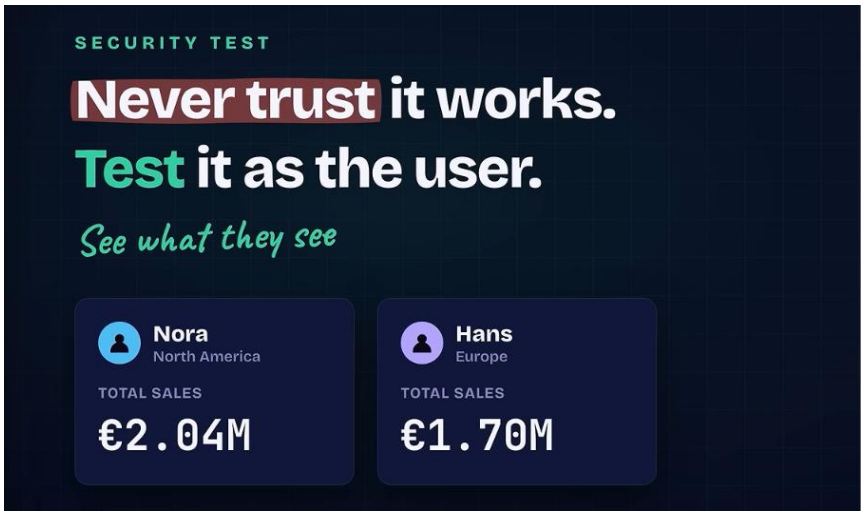


Step to create security

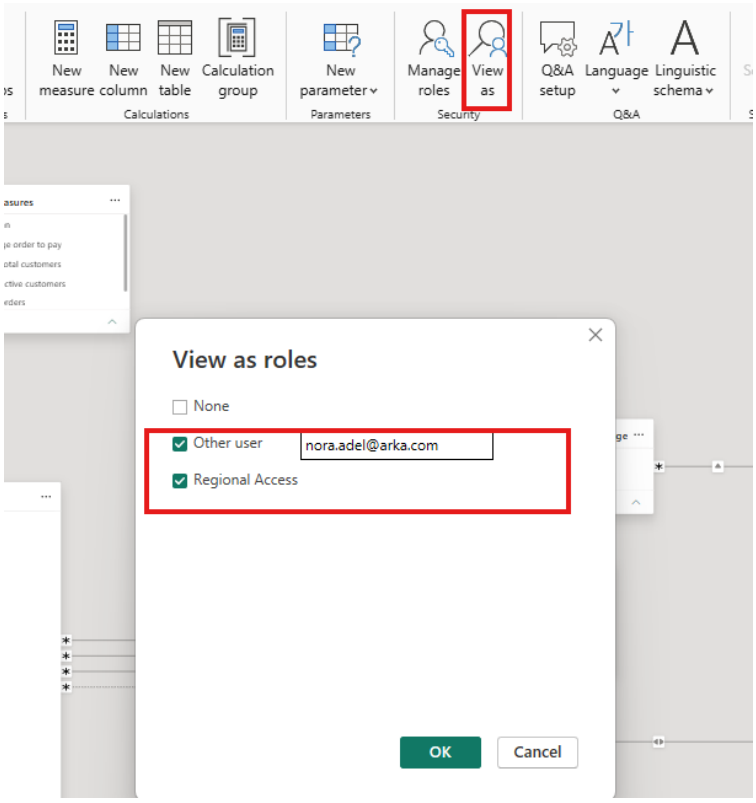
- First connect the security list to dim table based on requirement
- Go to manager role
- Click on new and give name
- Tables = choose on which table we are giving security level
- Write dax for user login filtration


Manage security roles
Create new security roles and use filters to define row-level data restrictions.
Roles
New
Regional Access
Tables
_measures
dim_campaign
dim_customer
dim_date
dim_geo
dim_order_flags
dim_product
fact_campaign...
fact_inventory
fact_order_pro...
fact_promotio...
Rules
Switch to default editor
1 [region_name] = LOOKUPVALUE(security[region],security[user_email],
USERPRINCIPALNAME())
Filter the data that this role can see by entering a DAX filter expression that returns a True/False value. For example: [Entity ID] = "Value"
Save
Close

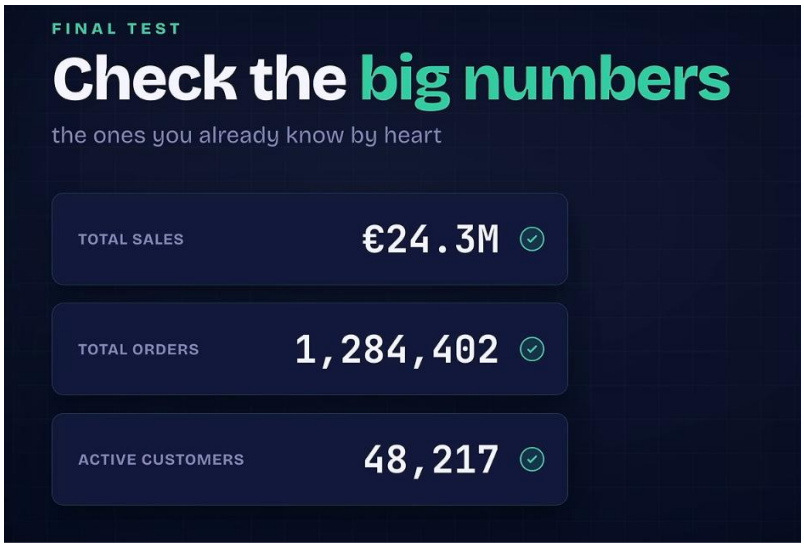

Create new security roles and use filters to define row-level data restrictions.
Successfully applied role changes.
Roles
New
Regional Access
Tables
_measures
dim_campaign
dim_customer
dim_date
dim_geo
dim_order_flags
dim_product
fact_campaign...
fact_inventory
fact_order_pro...
fact_promotio...
Rules
Switch to default editor
1 [region_name] = LOOKUPVALUE(security[region],security[user_email],
USERPRINCIPALNAME())
Filter the data that this role can see by entering a DAX filter expression that returns a True/False value. For example: [Entity ID] = "Value"
Save
Close


SECURITY TEST
Never trust it works.
Test it as the user.
See what they see
Nora
North America
TOTAL SALES
€2.04M
Hans
Europe
TOTAL SALES
€1.70M

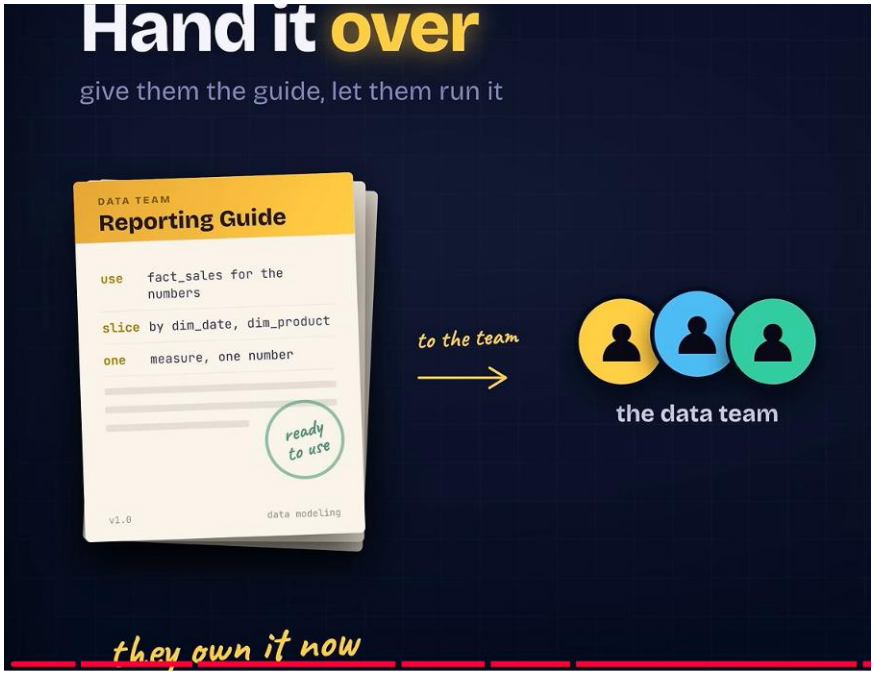
- *Copy email from list*
- *Go to View as*
- *Click on the rule we created and mention mail and click on other user as well.*

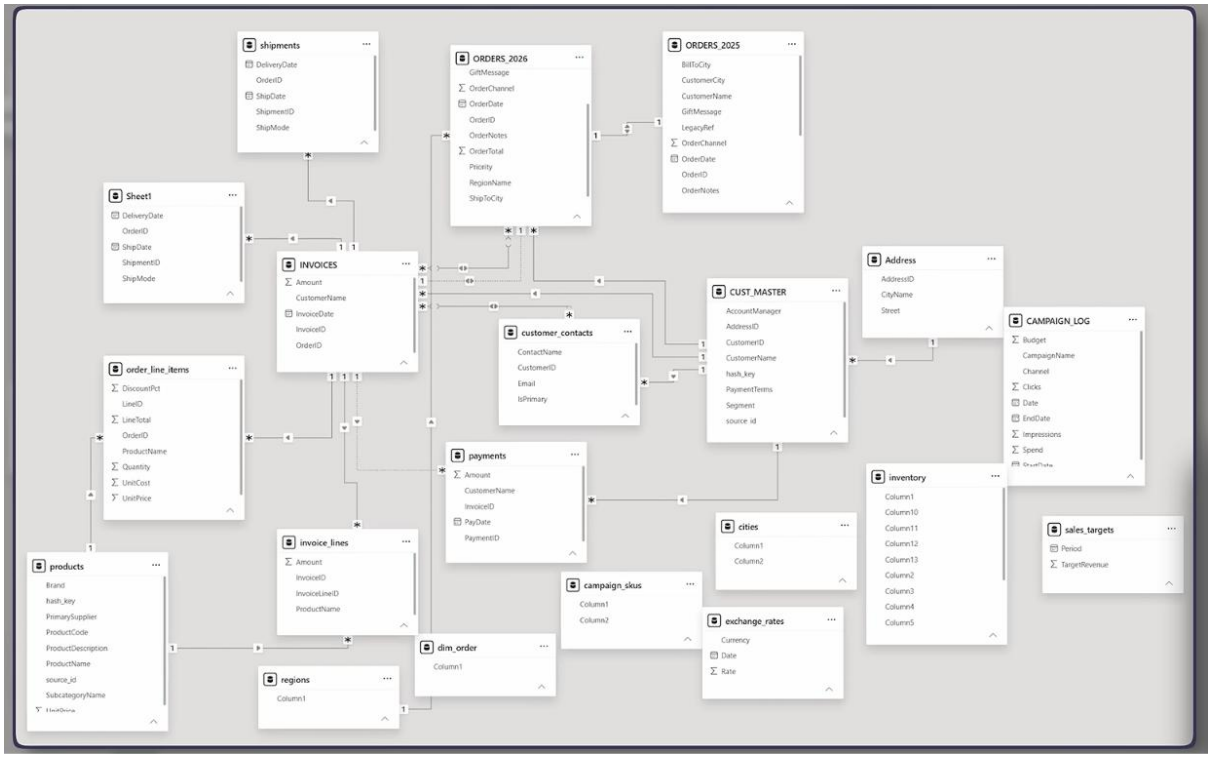


Step 5 – Final Test



To make sure that everything is correct before handing over to others.




shipments
DeliveryDate
OrderID
ShipDate
ShipmentID
ShipMode
ORDERS_2026
GiftMessage
OrderChannel
OrderDate
OrderID
OrderNotes
OrderTotal
Priority
RegionName
ShipToCity
ORDERS_2025
BillToCity
CustomerCity
CustomerName
GiftMessage
LegacyRef
OrderChannel
OrderDate
OrderID
OrderNotes
Sheet1
DeliveryDate
OrderID
ShipDate
ShipmentID
ShipMode
INVOICES
Amount
CustomerName
InvoiceDate
InvoiceID
OrderID
customer_contacts
ContactName
CustomerID
Email
IsPrimary
CUST_MASTER
AccountManager
AddressID
CustomerID
CustomerName
hash_key
PaymentTerms
Segment
source_id
Address
AddressID
CityName
Street
CAMPAIGN_LOG
Budget
CampaignName
Channel
Clicks
Date
EndDate
Impressions
Spend
order_line_items
DiscountPct
LineID
LineTotal
OrderID
ProductName
Quantity
UnitCost
UnitPrice
payments
Amount
CustomerName
InvoiceID
PayDate
PaymentID
inventory
Column1
Column10
Column11
Column12
Column13
Column2
Column3
Column4
Column5
cities
Column1
Column2
sales_targets
Period
TargetRevenue
products
Brand
hash_key
PrimarySupplier
ProductCode
ProductDescription
ProductName
source_id
SubcategoryName
invoice_lines
Amount
InvoiceID
InvoiceLineID
ProductName
campaign_skus
Column1
Column2
exchange_rates
Currency
Date
Rate
dim_order
Column1
regions
Column1